CLOUD COMPUTING

LAB NO# 06

SUBMITTED TO:

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Semester: V-A

CLOUD COMPUTING

LAB NO# 06

Task 1: Switch to root with su and back to a normal user

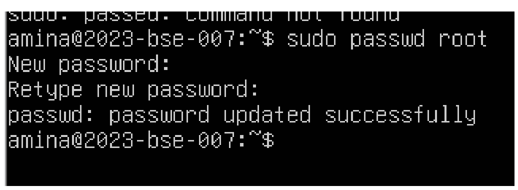
Step 1: Set a root password

Ubuntu disables the root account by default, so you need to assign it a password.

1. Open your terminal.
2. Run the command:

sudo passwd root

3. You will be prompted for your **current user password** (your normal account password).
4. Then it will ask you to **enter a new UNIX password**. This will be the root password for this lab.
5. Confirm the password by entering it again.



Step 2: Switch to the root account

Now, use su - to switch to the root user:

1. In the terminal, type:

su -

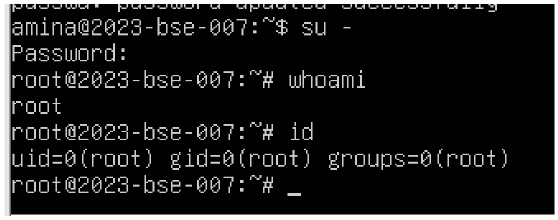
2. Enter the root password you just set.
3. Verify you are root by checking your username and ID:

whoami

id

- whoami should return root.

- id will show uid=0(root) among other information.



Step 3: Switch back to your normal user

To return to your regular account:

1. Type:

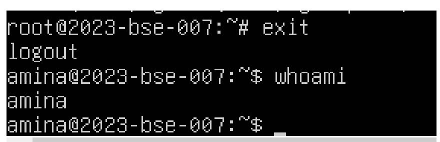
exit

or press Ctrl+D.

2. Verify you're back to your normal user:

whoami

- This should return your normal username (e.g., amina).



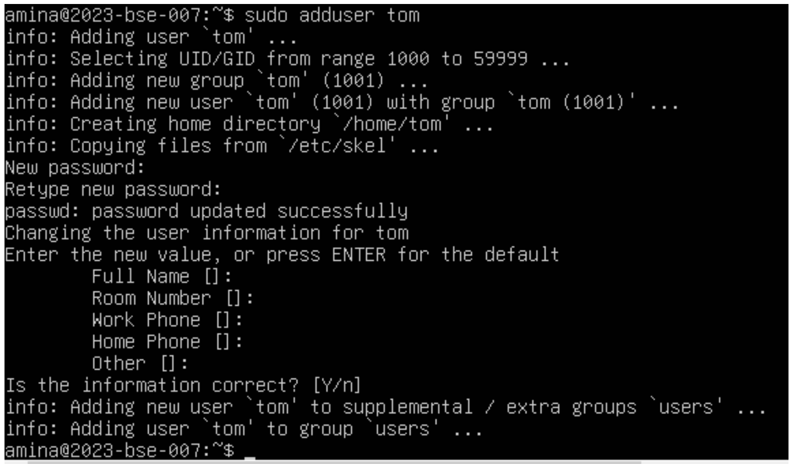
Task 2 – Create user tom and verify in passwd/group/shadow

Step 1: Create user tom

1. Open your terminal.
2. Run:

sudo adduser tom

3. The system will ask you to:
   - Enter a password for tom
   - Confirm the password
   - Optionally enter full name, room number, etc. (you can press **Enter** to skip)



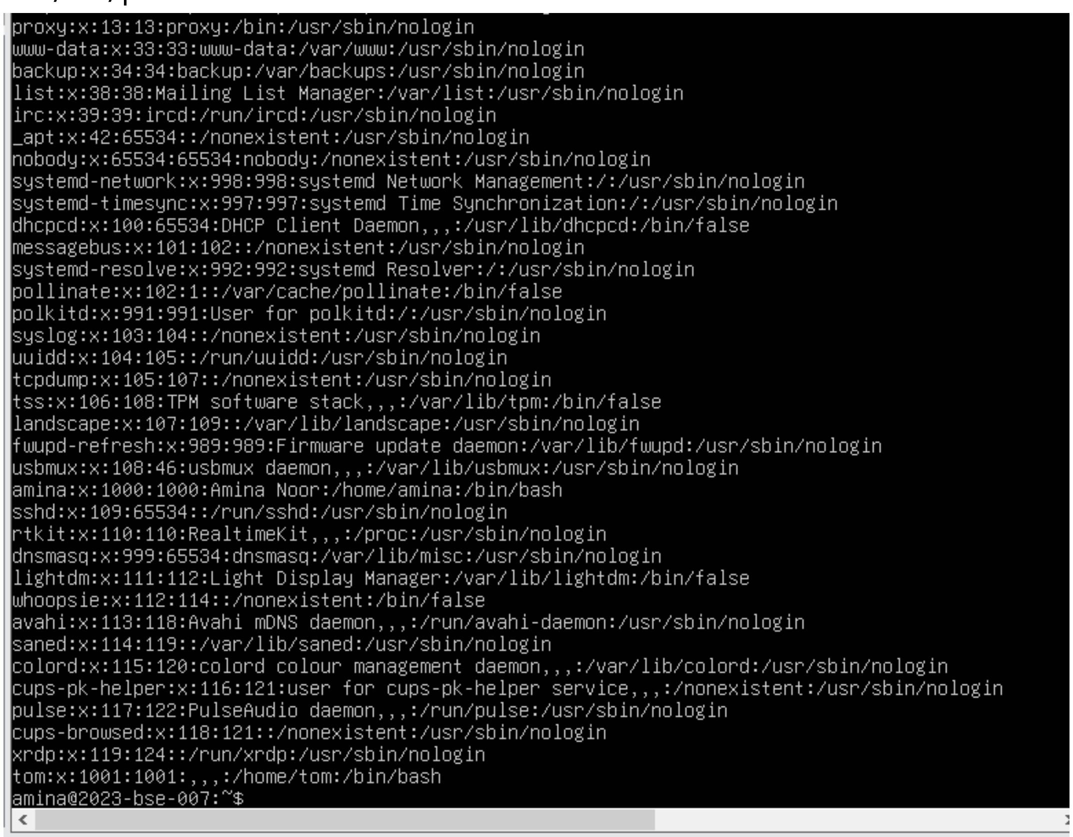
```
amina@2023-bse-007:~$ sudo adduser tom
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1001) ...
info: Adding new user `tom' (1001) with group `tom (1001)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
amina@2023-bse-007:~$ _
```

Step 2: Verify tom in /etc/passwd

/etc/passwd contains basic user info.

1. Run:

cat /etc/passwd



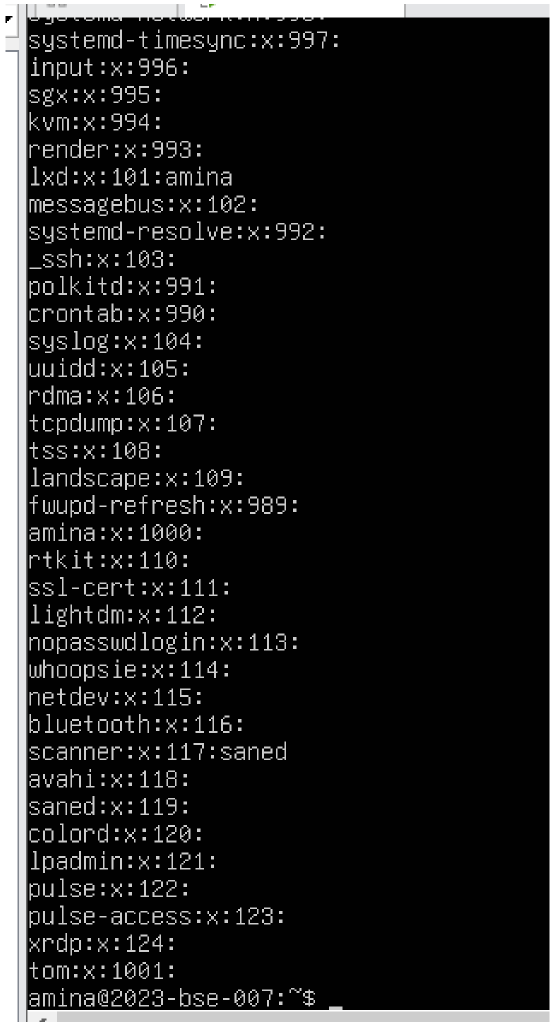
```
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
amina:x:1000:1000:Amina Noor:/home/amina:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
lightdm:x:111:112:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:112:114::/nonexistent:/bin/false
avahi:x:113:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:114:119::/var/lib/saned:/usr/sbin/nologin
colord:x:115:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
cups-pk-helper:x:116:121:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:121::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1001:,,,:/home/tom:/bin/bash
amina@2023-bse-007:~$
```

Step 3: Verify tom in /etc/group

/etc/group lists all groups on the system.

1. Run:

cat /etc/group



```
systemd-timesync:x:997:
input:x:996:
sgx:x:995:
kvm:x:994:
render:x:993:
lxd:x:101:amina
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uuidd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
amina:x:1000:
rtkit:x:110:
ssl-cert:x:111:
lightdm:x:112:
nopasswdlogin:x:113:
whoopsie:x:114:
netdev:x:115:
bluetooth:x:116:
scanner:x:117:saned
avahi:x:118:
saned:x:119:
colord:x:120:
lpadmin:x:121:
pulse:x:122:
pulse-access:x:123:
xrdp:x:124:
tom:x:1001:
amina@2023-bse-007:~$ _
```

Step 4: Verify tom in /etc/shadow

/etc/shadow stores password hashes and requires root access to view.

1. Run:

sudo cat /etc/shadow

Task 3: Create groups; change tom's primary and secondary groups

Step 1: Create groups

Create developer, devops, and designer groups:

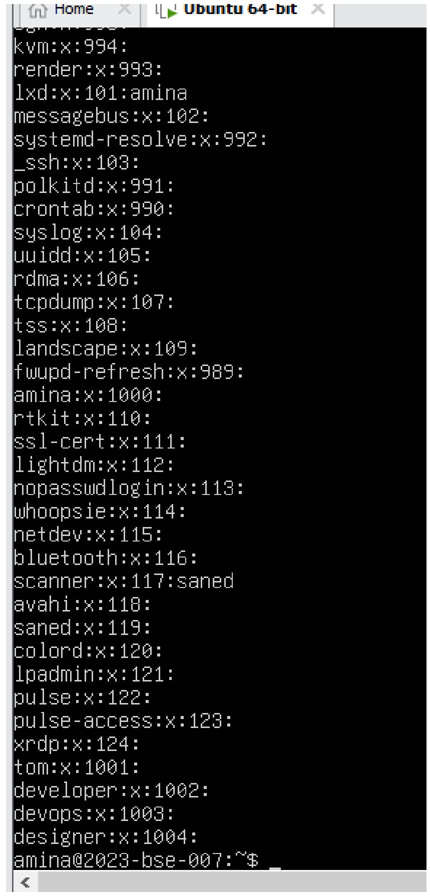
sudo groupadd developer

sudo groupadd devops

sudo groupadd designer

Verify they exist in /etc/group:

cat /etc/group



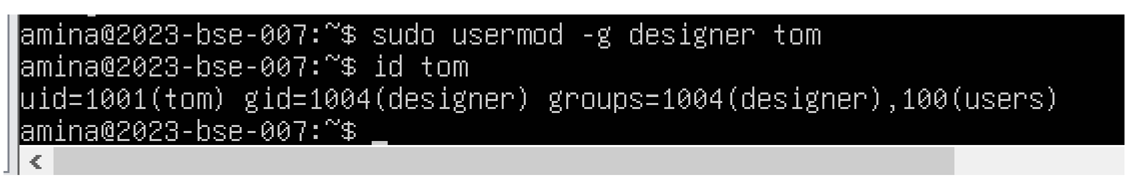
Step 2: Change tom's primary group

The primary group is the group listed in /etc/passwd as the user's main group. Change it to designer:

sudo usermod -g designer tom

Verify:

id tom



Step 3: Add secondary groups (developer and devops)

Add tom to multiple secondary groups **without removing existing groups**:

sudo usermod -aG developer,devops tom

Verify:

id tom

groups tom

```
amina@2023-bse-007:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
amina@2023-bse-007:~$ groups tom
tom : designer users
amina@2023-bse-007:~$
```

Step 4: Reset secondary groups

Replace all secondary groups so only tom's own group remains:

sudo usermod -G tom tom

Verify:

id tom

groups tom

```
amina@2023-bse-007:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
amina@2023-bse-007:~$ groups tom
tom : designer users
amina@2023-bse-007:~$ _
```

Task 4 – Create/delete users (Jerry, Scooby) and groups (jolly, anime)

Step 1: Create users

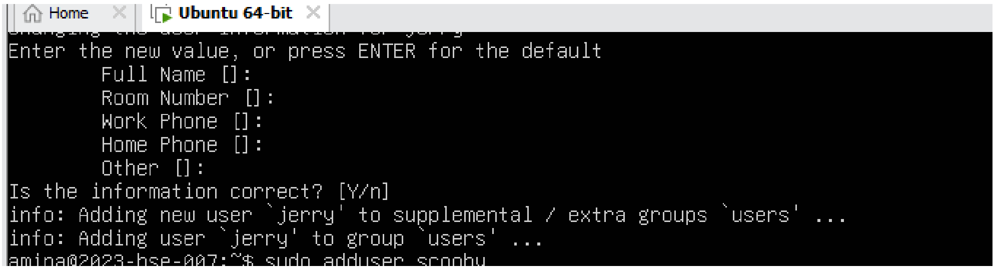
1. **Create Jerry using adduser (interactive, creates home & password):**

sudo adduser Jerry

- Follow prompts for password and optional info.

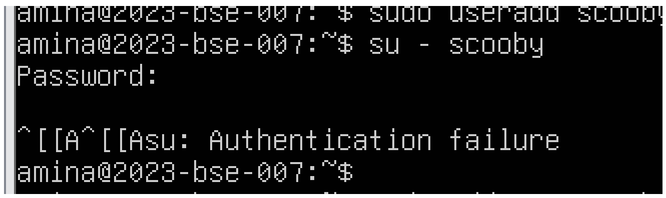
2. **Create Scooby using useradd (non-interactive, no home, no password):**

sudo useradd Scooby



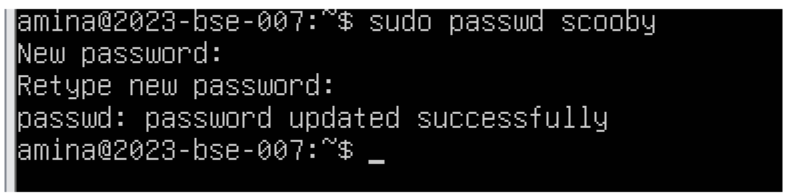
Step 2: Try logging in as Scooby

su - Scooby



Step 3: Set a password for Scooby

sudo passwd Scooby



Step 4: Try logging in as Scooby again (home directory missing)

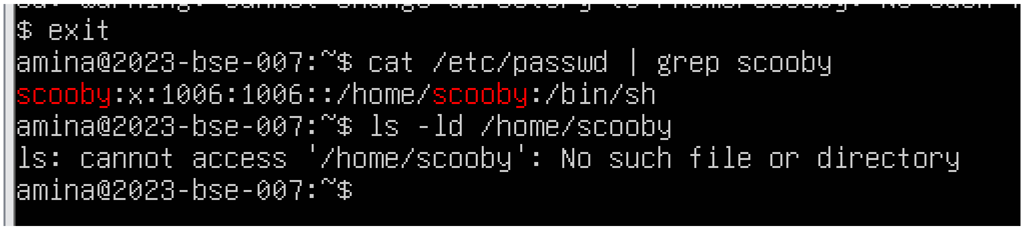
su - Scooby

Step 5: Show Scooby's home directory doesn't exist

exit

cat /etc/passwd | grep Scooby

ls -ld /home/Scooby



```
$ exit
amina@2023-bse-007:~$ cat /etc/passwd | grep scooby
scooby:x:1006:1006::/home/scooby:/bin/sh
amina@2023-bse-007:~$ ls -ld /home/scooby
ls: cannot access '/home/scooby': No such file or directory
amina@2023-bse-007:~$
```

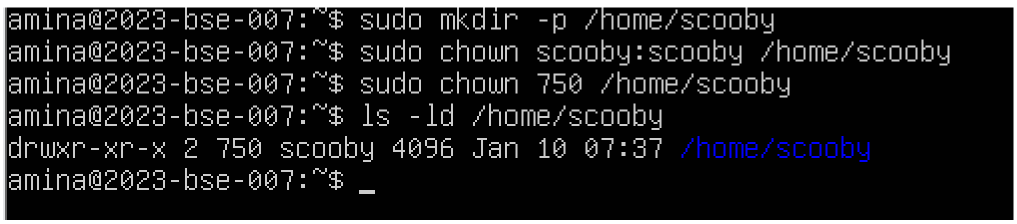
Step 6: Create Scooby's home directory and set permissions

sudo mkdir -p /home/Scooby

sudo chown Scooby:Scooby /home/Scooby

sudo chmod 750 /home/Scooby

ls -ld /home/Scooby



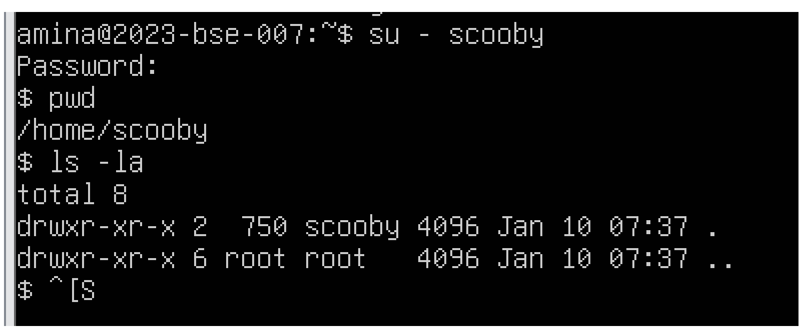
```
amina@2023-bse-007:~$ sudo mkdir -p /home/scooby
amina@2023-bse-007:~$ sudo chown scooby:scooby /home/scooby
amina@2023-bse-007:~$ sudo chown 750 /home/scooby
amina@2023-bse-007:~$ ls -ld /home/scooby
drwxr-xr-x 2 750 scooby 4096 Jan 10 07:37 /home/scooby
amina@2023-bse-007:~$ _
```

Step 7: Log in as Scooby and verify home directory

su - Scooby

pwd

ls -la

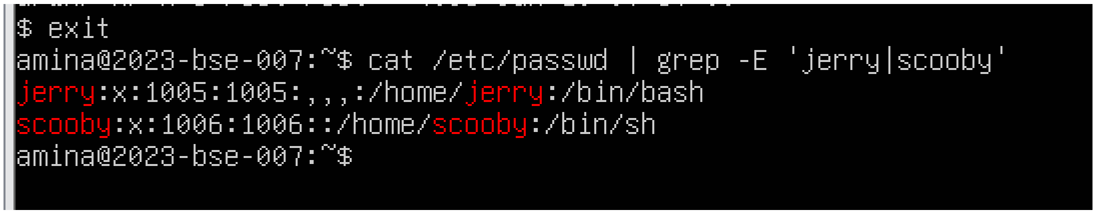


```
amina@2023-bse-007:~$ su - scooby
Password:
$ pwd
/home/scooby
$ ls -la
total 8
drwxr-xr-x 2  750 scooby 4096 Jan 10 07:37 .
drwxr-xr-x 6 root root   4096 Jan 10 07:37 ..
$ ^[S
```

Step 8: Verify users and observe Scooby's shell

exit

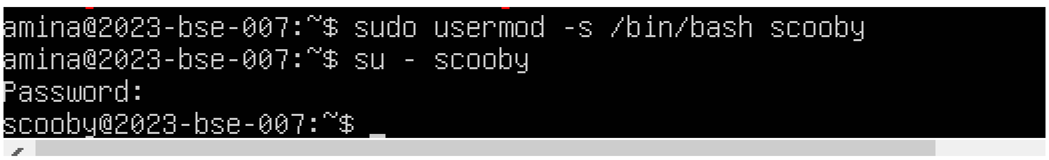
cat /etc/passwd | grep -E 'Jerry|Scooby'



Step 9: Change Scooby's shell to /bin/bash

sudo usermod -s /bin/bash Scooby

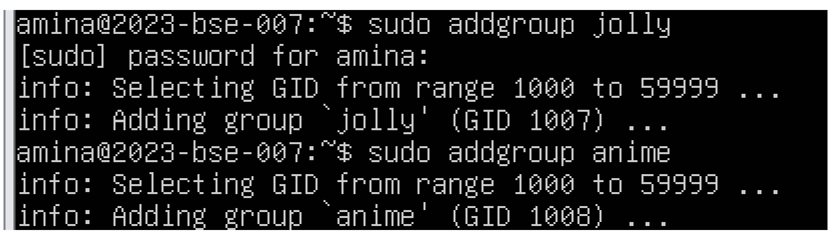
su - Scooby



Step 10: Create groups

sudo addgroup jolly # interactive

sudo groupadd anime # non-interactive



Step 11: Verify groups

cat /etc/group | grep -E 'jolly|anime'

Step 12: Delete groups

sudo delgroup jolly

sudo groupdel anime

cat /etc/group | grep -E 'jolly|anime'

```
amina@2023-bse-007:~$ sudo delgroup jolly
info: Removing group `jolly' ...
amina@2023-bse-007:~$ sudo delgroup anime
info: Removing group `anime' ...
amina@2023-bse-007:~$ cat /etc/group | grep -E 'jolly|anime'
```

Step 13: Delete users

sudo deluser --remove-home Jerry

sudo userdel -r Scooby

cat /etc/passwd | grep -E 'Jerry|Scooby'

```
amina@2023-bse-007:~$ sudo deluser --remove-home jerry
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `jerry' ...
amina@2023-bse-007:~$ sudo userdel -r scooby
userdel: scooby mail spool (/var/mail/scooby) not found
userdel: /home/scooby not owned by scooby, not removing
amina@2023-bse-007:~$ cat /etc/passwd | grep -E 'jolly|anime'
amina@2023-bse-007:~$ _
```

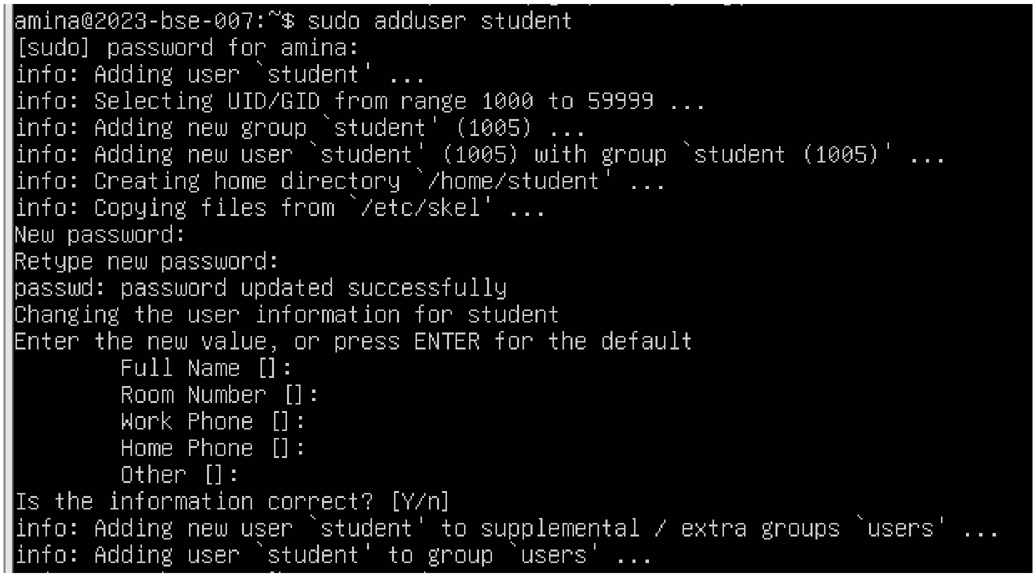
Task 5: Create user Student; create files; set owner/group; identify file types

Create user Student

Open terminal and run:

sudo adduser Student

- Set password
- Other details → press **ENTER** to skip



```
amina@2023-bse-007:~$ sudo adduser student
[sudo] password for amina:
info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1005) ...
info: Adding new user `student' (1005) with group `student (1005)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...
```

2 Switch to Student & create files

su - Student

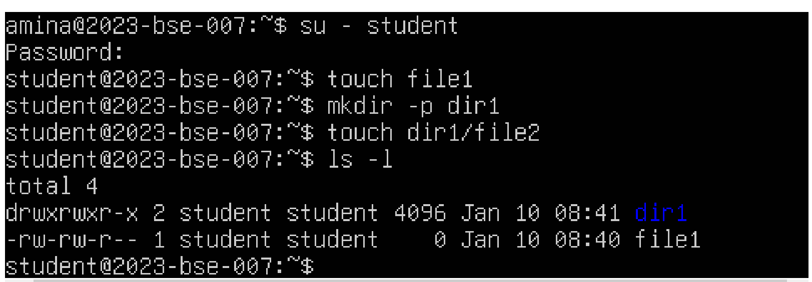
Now create files and directory:

touch file1

mkdir -p dir1

touch dir1/file2

ls -l



```
amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ touch file1
student@2023-bse-007:~$ mkdir -p dir1
student@2023-bse-007:~$ touch dir1/file2
student@2023-bse-007:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Jan 10 08:41 dir1
-rw-rw-r-- 1 student student    0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

Change owner of file1

⚠ This must be done using **sudo**, so it works only if:

- user **tom** already exists
- group **devops** already exists

If they don't exist, create them first:

exit

sudo adduser tom

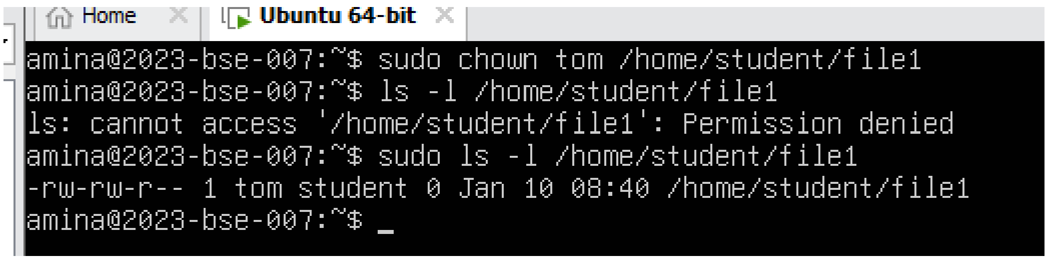
sudo addgroup devops

su - Student

Now change owner:

sudo chown tom file1

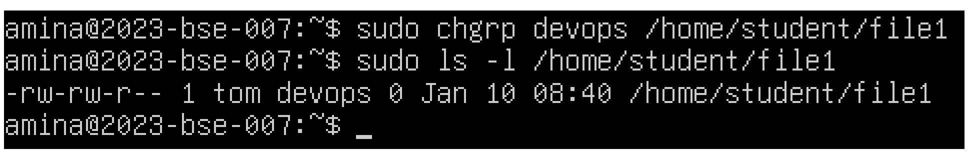
ls -l file1



Change group of file1

sudo chgrp devops file1

ls -l file1



Identify file types & show /dev/null

Run these commands:

ls -l

ls -l dir1

ls -l /dev/null

file file1 dir1 /dev/null

```
amina@2023-bse-007:~$ sudo ls -l /home/student
total 4
drwxrwxr-x 2 student student 4096 Jan 10 08:41 dir1
-rw-rw-r-- 1 tom     devops     0 Jan 10 08:40 file1
amina@2023-bse-007:~$ sudo ls -l /home/student/dir1
total 0
-rw-rw-r-- 1 student student 0 Jan 10 08:41 file2
amina@2023-bse-007:~$ sudo ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Jan 10 07:02 /dev/null
amina@2023-bse-007:~$ sudo file /home/student/file1 /home/student/dir1 /dev/null
/home/student/file1: empty
/home/student/dir1:  directory
/dev/null:           character special (1/3)
amina@2023-bse-007:~$
```

Exit Student account

Exit

```
amina@2023-bse-007:~$ _
```

Task 6: Change permissions using symbolic mode

Switch to Student & verify file

su - Student

cd ~

ls -l file1

```
amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ cd ~
student@2023-bse-007:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Jan 10 08:40 file1
```

Remove ALL permissions from file1

chmod a-rwx file1

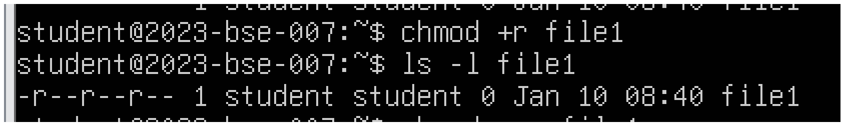
ls -l file1

```
student@2023-bse-007:~$ chmod a-rwx file1
student@2023-bse-007:~$ ls -l file1
---------- 1 student student 0 Jan 10 08:40 file1
```

Add READ permission to ALL

chmod +r file1

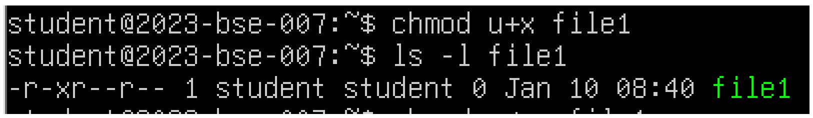
ls -l file1


```
student@2023-bse-007:~$ chmod +r file1
student@2023-bse-007:~$ ls -l file1
-r--r--r-- 1 student student 0 Jan 10 08:40 file1
```

Add EXECUTE permission to USER only

chmod u+x file1

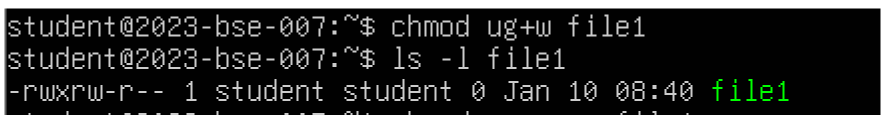
ls -l file1


```
student@2023-bse-007:~$ chmod u+x file1
student@2023-bse-007:~$ ls -l file1
-r-xr--r-- 1 student student 0 Jan 10 08:40 file1
```

Add WRITE permission to USER and GROUP

chmod ug+w file1

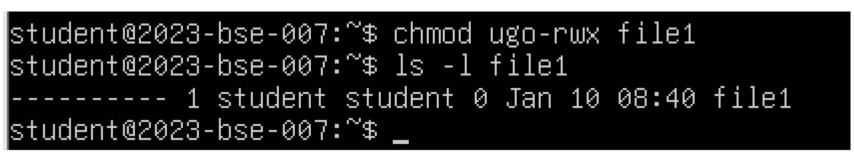
ls -l file1


```
student@2023-bse-007:~$ chmod ug+w file1
student@2023-bse-007:~$ ls -l file1
-rwxrw-r-- 1 student student 0 Jan 10 08:40 file1
```

Remove ALL permissions explicitly

chmod ugo-rwx file1

ls -l file1


```
student@2023-bse-007:~$ chmod ugo-rwx file1
student@2023-bse-007:~$ ls -l file1
---------- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _
```

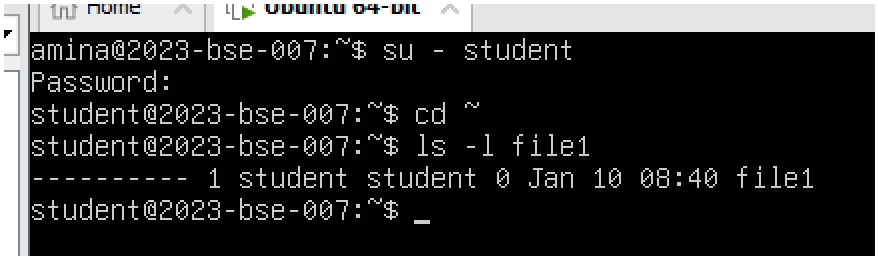
Task 7 – Change Permissions using SET (=) Symbolic Mode

Ensure Student context

su - Student

cd ~

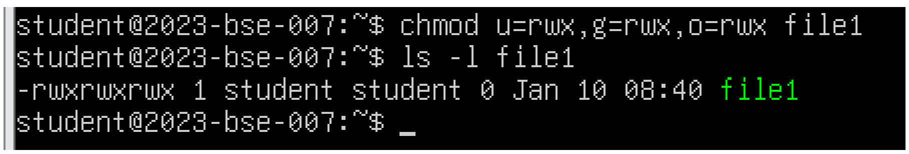
ls -l file1



Set ALL permissions to rwx

chmod u=rwx,g=rwx,o=rwx file1

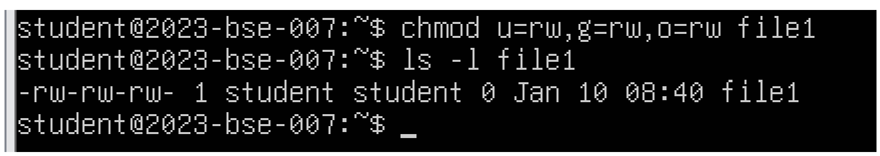
ls -l file1



Remove execute from GROUP and OTHERS

chmod g=rw,o=rw file1

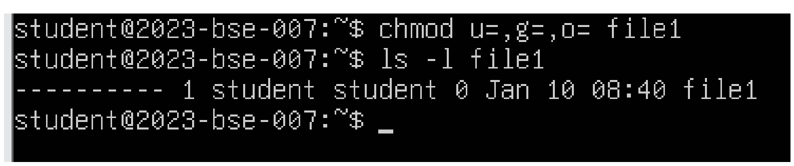
ls -l file1



Remove ALL permissions from everyone

chmod u=,g=,o= file1

ls -l file1



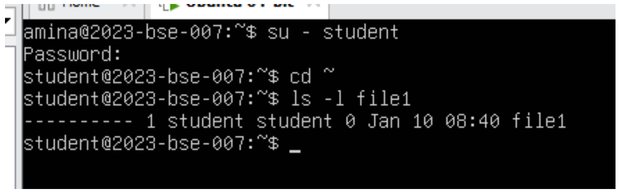
Task 8 – Change Permissions using Numeric (Octal) Mode

Ensure Student context

su - Student

cd ~

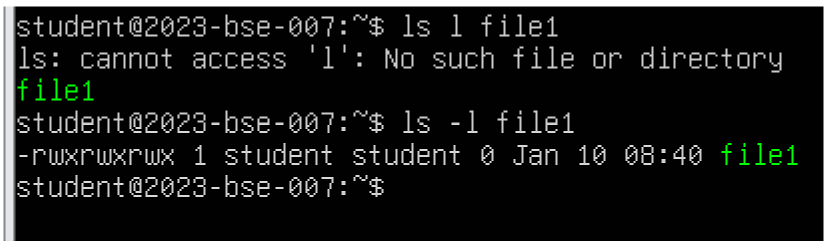
ls -l file1



chmod 777 → full access to everyone

chmod 777 file1

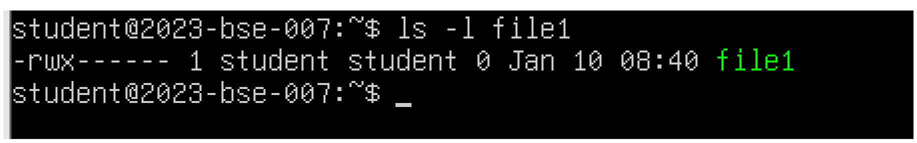
ls -l file1



chmod 700 → only owner has access

chmod 700 file1

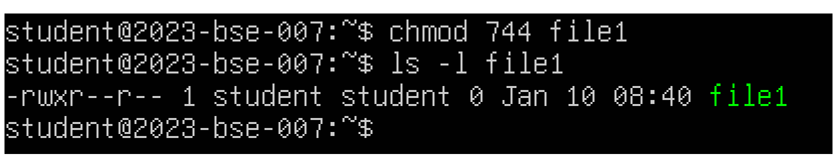
ls -l file1



chmod 744 → owner full, others read

chmod 744 file1

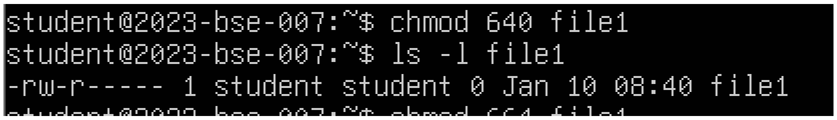
ls -l file1



chmod 640 → owner read/write, group read

chmod 640 file1

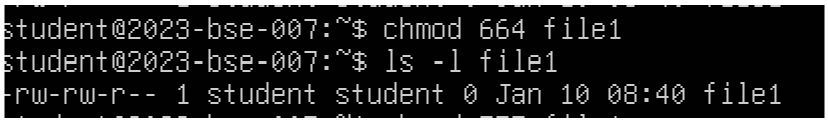
ls -l file1


```
student@2023-bse-007:~$ chmod 640 file1
student@2023-bse-007:~$ ls -l file1
-rw-r----- 1 student student 0 Jan 10 08:40 file1
```

chmod 664 → owner/group rw, others read

chmod 664 file1

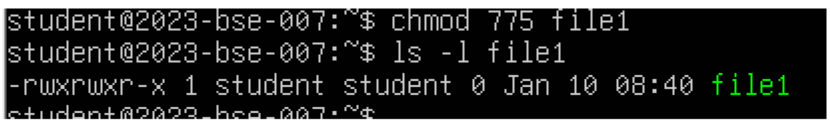
ls -l file1


```
student@2023-bse-007:~$ chmod 664 file1
student@2023-bse-007:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Jan 10 08:40 file1
```

chmod 775 → owner/group full, others rx

chmod 775 file1

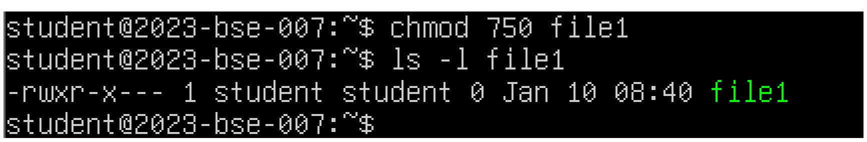
ls -l file1


```
student@2023-bse-007:~$ chmod 775 file1
student@2023-bse-007:~$ ls -l file1
-rwxrwxr-x 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

chmod 750 → owner full, group rx

chmod 750 file1

ls -l file1


```
student@2023-bse-007:~$ chmod 750 file1
student@2023-bse-007:~$ ls -l file1
-rwxr-x--- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

Task 9 – Pipes, Pagers, Grep & Redirects

View syslog using less

sudo cat /var/log/syslog | less

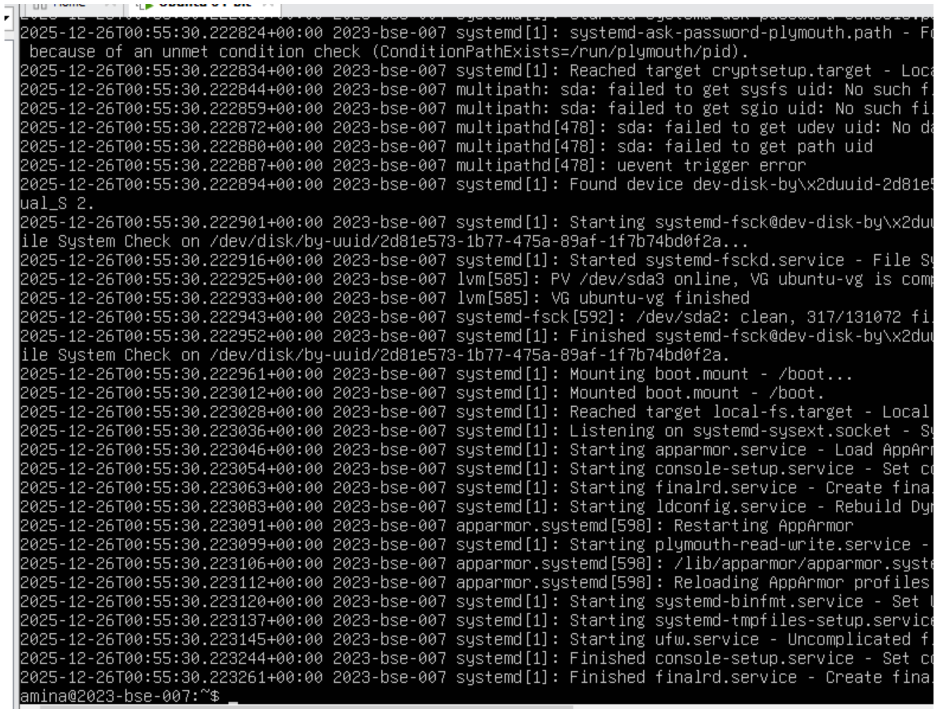
- Scroll with **↑↓ / PageUp / PageDown**
- Quit with **q**

```
2025-12-26T00:55:30.220400+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x0000000000100000-0x000000
2025-12-26T00:55:30.220401+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x000000007fee0000-0x000000
2025-12-26T00:55:30.220401+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x000000007feff000-0x000000
2025-12-26T00:55:30.220402+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x000000007ff00000-0x000000
2025-12-26T00:55:30.220403+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000f0000000-0x000000
2025-12-26T00:55:30.220403+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fec00000-0x000000
2025-12-26T00:55:30.220404+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fee00000-0x000000
2025-12-26T00:55:30.220411+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fffe0000-0x000000
2025-12-26T00:55:30.220412+00:00 2023-bse-007 kernel: NX (Execute Disable) protection: active
2025-12-26T00:55:30.220413+00:00 2023-bse-007 kernel: APIC: Static calls initialized
2025-12-26T00:55:30.220413+00:00 2023-bse-007 kernel: SMBIOS 2.7 present.
2025-12-26T00:55:30.220414+00:00 2023-bse-007 kernel: DMI: VMware, Inc. VMware Virtual Platform/4
2025-12-26T00:55:30.220415+00:00 2023-bse-007 kernel: vmware: hypercall mode: 0x02
2025-12-26T00:55:30.220421+00:00 2023-bse-007 kernel: Hypervisor detected: VMware
2025-12-26T00:55:30.220422+00:00 2023-bse-007 kernel: vmware: TSC freq read from hypervisor : 249
2025-12-26T00:55:30.220423+00:00 2023-bse-007 kernel: vmware: Host bus clock speed read from hype
2025-12-26T00:55:30.220423+00:00 2023-bse-007 kernel: vmware: using clock offset of 4084524726 ns
2025-12-26T00:55:30.220424+00:00 2023-bse-007 kernel: tsc: Detected 2496.000 MHz processor
2025-12-26T00:55:30.220425+00:00 2023-bse-007 kernel: e820: update [mem 0x00000000-0x00000fff] us
2025-12-26T00:55:30.220431+00:00 2023-bse-007 kernel: e820: remove [mem 0x000a0000-0x000fffff] us
2025-12-26T00:55:30.220433+00:00 2023-bse-007 kernel: last_pfn = 0x80000 max_arch_pfn = 0x4000000
2025-12-26T00:55:30.220434+00:00 2023-bse-007 kernel: total RAM covered: 3072M
2025-12-26T00:55:30.220435+00:00 2023-bse-007 kernel: Found optimal setting for mtrr clean up
2025-12-26T00:55:30.220436+00:00 2023-bse-007 kernel:  gran_size: 64K   chunk_size: 64K        n
2025-12-26T00:55:30.220437+00:00 2023-bse-007 kernel: MTRR map: 6 entries (5 fixed + 1 variable;
2025-12-26T00:55:30.220438+00:00 2023-bse-007 kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- U
2025-12-26T00:55:30.222460+00:00 2023-bse-007 systemd-modules-load[462]: Inserted module 'msr'
2025-12-26T00:55:30.222586+00:00 2023-bse-007 systemd-modules-load[462]: Inserted module 'dm_mult
2025-12-26T00:55:30.222611+00:00 2023-bse-007 lvm[445]:   1 logical volume(s) in volume group "ub
2025-12-26T00:55:30.222622+00:00 2023-bse-007 systemd[1]: Finished systemd-sysctl.service - Apply
2025-12-26T00:55:30.222631+00:00 2023-bse-007 systemd[1]: Finished systemd-tmpfiles-setup-dev-ear
2025-12-26T00:55:30.222642+00:00 2023-bse-007 systemd[1]: Starting systemd-journal-flush.service
2025-12-26T00:55:30.222651+00:00 2023-bse-007 multipathd[478]: multipathd v0.9.4: start up
2025-12-26T00:55:30.222659+00:00 2023-bse-007 systemd[1]: Starting systemd-sysusers.service - Cre
2025-12-26T00:55:30.222676+00:00 2023-bse-007 multipathd[478]: reconfigure: setting up paths and
amina@2023-bse-007:~$ _
```

View syslog using more

sudo cat /var/log/syslog | more

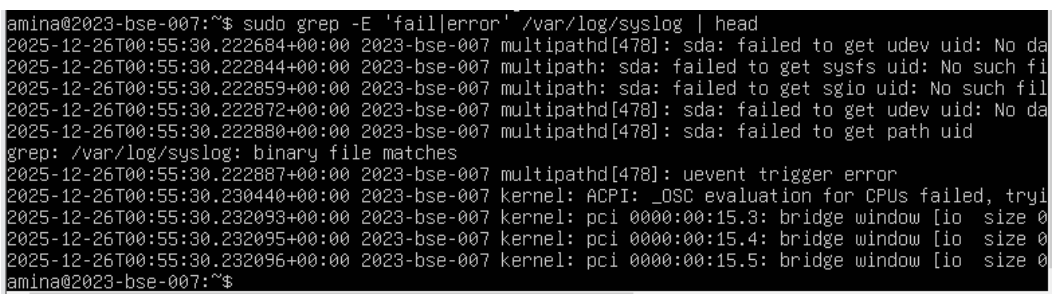
- Press **SPACE** for next page
- Press **q** to quit



```
2025-12-26T00:55:30.222824+00:00 2023-bse-007 systemd[1]: systemd-ask-password-plymouth.path - Fo
 because of an unmet condition check (ConditionPathExists=/run/plymouth/pid).
2025-12-26T00:55:30.222834+00:00 2023-bse-007 systemd[1]: Reached target cryptsetup.target - Loca
2025-12-26T00:55:30.222844+00:00 2023-bse-007 multipath: sda: failed to get sysfs uid: No such f
2025-12-26T00:55:30.222859+00:00 2023-bse-007 multipath: sda: failed to get sgio uid: No such fi
2025-12-26T00:55:30.222872+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No da
2025-12-26T00:55:30.222880+00:00 2023-bse-007 multipathd[478]: sda: failed to get path uid
2025-12-26T00:55:30.222887+00:00 2023-bse-007 multipathd[478]: uevent trigger error
2025-12-26T00:55:30.222894+00:00 2023-bse-007 systemd[1]: Found device dev-disk-by\x2duuid-2d81e5
ual_S 2.
2025-12-26T00:55:30.222901+00:00 2023-bse-007 systemd[1]: Starting systemd-fsck@dev-disk-by\x2duu
ile System Check on /dev/disk/by-uuid/2d81e573-1b77-475a-89af-1f7b74bd0f2a...
2025-12-26T00:55:30.222916+00:00 2023-bse-007 systemd[1]: Started systemd-fsckd.service - File Sy
2025-12-26T00:55:30.222925+00:00 2023-bse-007 lvm[585]: PV /dev/sda3 online, VG ubuntu-vg is comp
2025-12-26T00:55:30.222933+00:00 2023-bse-007 lvm[585]: VG ubuntu-vg finished
2025-12-26T00:55:30.222943+00:00 2023-bse-007 systemd-fsck[592]: /dev/sda2: clean, 317/131072 fi
2025-12-26T00:55:30.222952+00:00 2023-bse-007 systemd[1]: Finished systemd-fsck@dev-disk-by\x2duu
ile System Check on /dev/disk/by-uuid/2d81e573-1b77-475a-89af-1f7b74bd0f2a.
2025-12-26T00:55:30.222961+00:00 2023-bse-007 systemd[1]: Mounting boot.mount - /boot...
2025-12-26T00:55:30.223012+00:00 2023-bse-007 systemd[1]: Mounted boot.mount - /boot.
2025-12-26T00:55:30.223028+00:00 2023-bse-007 systemd[1]: Reached target local-fs.target - Local
2025-12-26T00:55:30.223036+00:00 2023-bse-007 systemd[1]: Listening on systemd-sysext.socket - Sy
2025-12-26T00:55:30.223046+00:00 2023-bse-007 systemd[1]: Starting apparmor.service - Load AppArm
2025-12-26T00:55:30.223054+00:00 2023-bse-007 systemd[1]: Starting console-setup.service - Set co
2025-12-26T00:55:30.223063+00:00 2023-bse-007 systemd[1]: Starting finalrd.service - Create fina
2025-12-26T00:55:30.223083+00:00 2023-bse-007 systemd[1]: Starting ldconfig.service - Rebuild Dyn
2025-12-26T00:55:30.223091+00:00 2023-bse-007 apparmor.systemd[598]: Restarting AppArmor
2025-12-26T00:55:30.223099+00:00 2023-bse-007 systemd[1]: Starting plymouth-read-write.service -
2025-12-26T00:55:30.223106+00:00 2023-bse-007 apparmor.systemd[598]: /lib/apparmor/apparmor.syste
2025-12-26T00:55:30.223112+00:00 2023-bse-007 apparmor.systemd[598]: Reloading AppArmor profiles
2025-12-26T00:55:30.223120+00:00 2023-bse-007 systemd[1]: Starting systemd-binfmt.service - Set U
2025-12-26T00:55:30.223137+00:00 2023-bse-007 systemd[1]: Starting systemd-tmpfiles-setup.service
2025-12-26T00:55:30.223145+00:00 2023-bse-007 systemd[1]: Starting ufw.service - Uncomplicated fi
2025-12-26T00:55:30.223244+00:00 2023-bse-007 systemd[1]: Finished console-setup.service - Set co
2025-12-26T00:55:30.223261+00:00 2023-bse-007 systemd[1]: Finished finalrd.service - Create fina
amina@2023-bse-007:~$ _
```

Grep failures/errors + head

sudo grep -E 'fail|error' /var/log/syslog | head

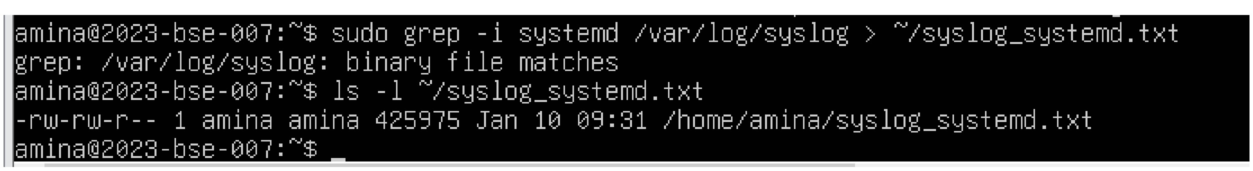


```
amina@2023-bse-007:~$ sudo grep -E 'fail|error' /var/log/syslog | head
2025-12-26T00:55:30.222684+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No da
2025-12-26T00:55:30.222844+00:00 2023-bse-007 multipath: sda: failed to get sysfs uid: No such fi
2025-12-26T00:55:30.222859+00:00 2023-bse-007 multipath: sda: failed to get sgio uid: No such fil
2025-12-26T00:55:30.222872+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No da
2025-12-26T00:55:30.222880+00:00 2023-bse-007 multipathd[478]: sda: failed to get path uid
grep: /var/log/syslog: binary file matches
2025-12-26T00:55:30.222887+00:00 2023-bse-007 multipathd[478]: uevent trigger error
2025-12-26T00:55:30.230440+00:00 2023-bse-007 kernel: ACPI: _OSC evaluation for CPUs failed, tryi
2025-12-26T00:55:30.232093+00:00 2023-bse-007 kernel: pci 0000:00:15.3: bridge window [io  size 0
2025-12-26T00:55:30.232095+00:00 2023-bse-007 kernel: pci 0000:00:15.4: bridge window [io  size 0
2025-12-26T00:55:30.232096+00:00 2023-bse-007 kernel: pci 0000:00:15.5: bridge window [io  size 0
amina@2023-bse-007:~$
```

Redirect output (overwrite >)

sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt

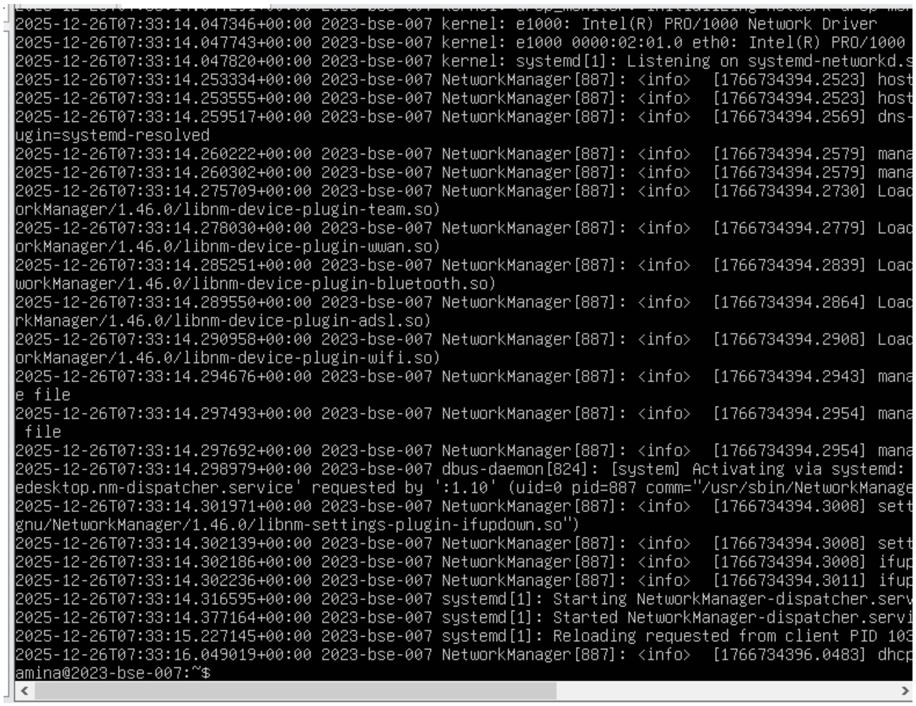
ls -l ~/syslog_systemd.txt



```
amina@2023-bse-007:~$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
amina@2023-bse-007:~$ ls -l ~/syslog_systemd.txt
-rw-rw-r-- 1 amina amina 425975 Jan 10 09:31 /home/amina/syslog_systemd.txt
amina@2023-bse-007:~$ _
```

Append output (>>)

sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt cat ~/syslog_systemd.txt



If /var/log/syslog does NOT exist (Very Common)

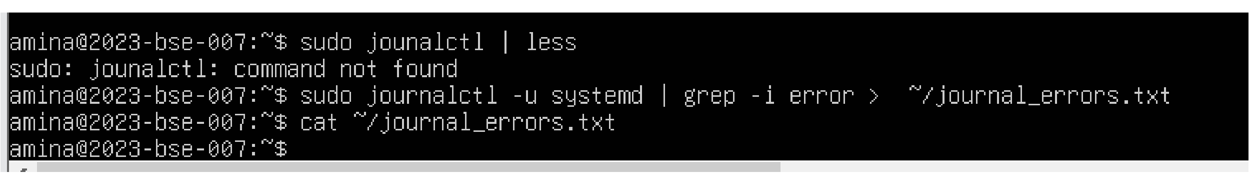
journalctl (alternative)

sudo journalctl | less

Filter journal logs and redirect

sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt

cat ~/journal_errors.txt



Task 10: Script setup.sh – variables, command substitution, file/dir checks, permissions (use vim)

Step 1: Create script with shebang

1. Open vim:

   vim setup.sh

2. Add the first line:

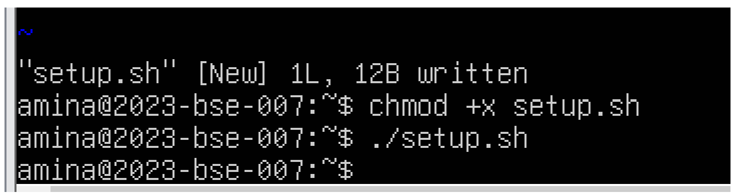
   #!/bin/bash

3. Save & quit :wq
4. Make executable:

   chmod +x setup.sh

5. Run it:

   ./setup.sh



Step 2: Define a variable and echo it

1. Open vim to append:

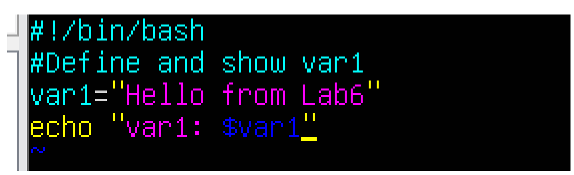
   vim setup.sh

2. Add:

   # Define and show var1

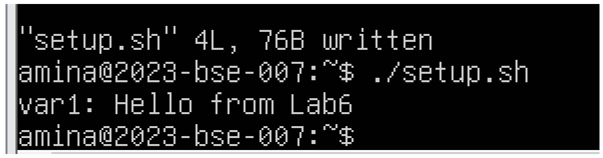
   var1="Hello from Lab 6"

   echo "var1: $var1"



3. Save & quit :wq
4. Run:

   ./setup.sh



Step 3: Save ls -l output into a variable

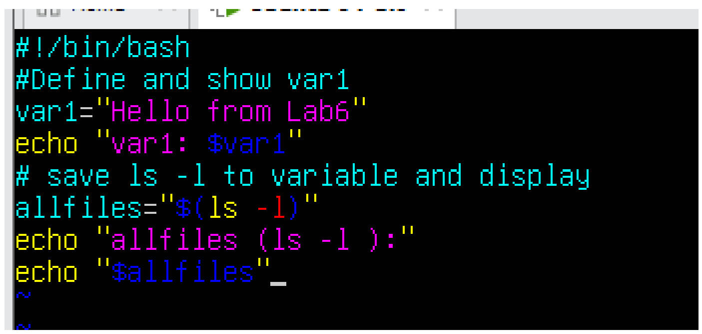
1. Vim appends:

   # Save ls -l to variable and display

   allFiles="$(ls -l)"

   echo "allFiles (ls -l):"

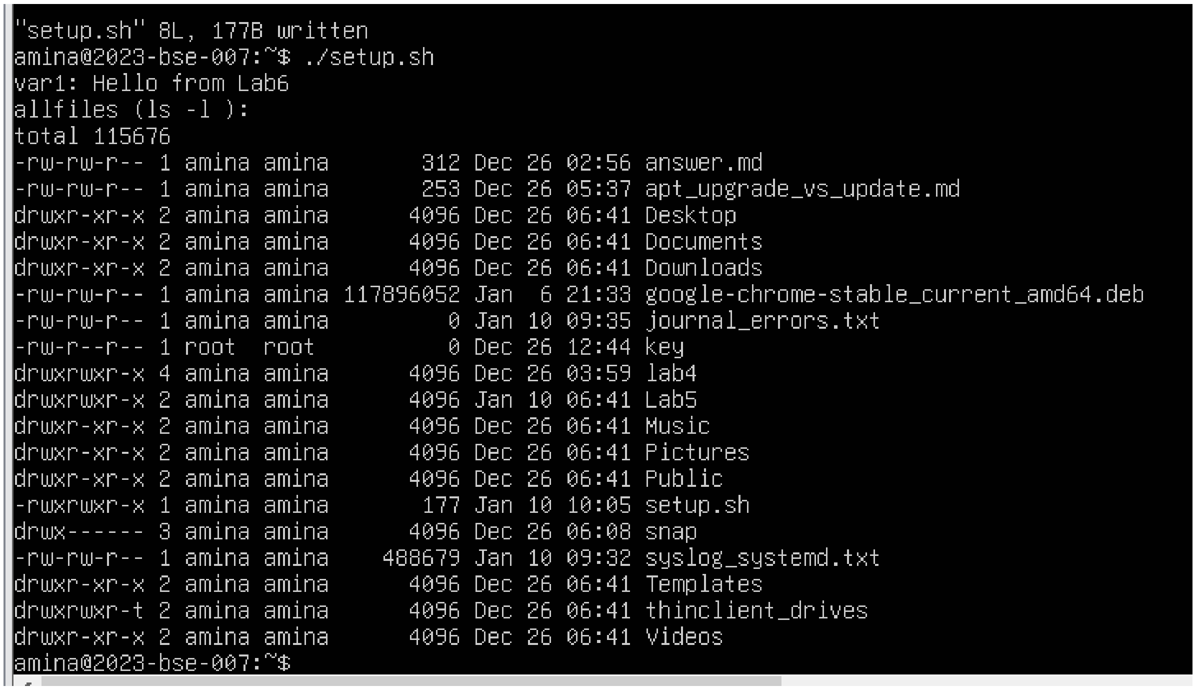
   echo "$allFiles"



2. Save & quit

3. Run:

   ./setup.sh



```
"setup.sh" 8L, 177B written
amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l ):
total 115676
-rw-rw-r-- 1 amina amina       312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina       253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Desktop
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina         0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root          0 Dec 26 12:44 key
drwxrwxr-x 4 amina amina      4096 Dec 26 03:59 lab4
drwxrwxr-x 2 amina amina      4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina       177 Jan 10 10:05 setup.sh
drwx------ 3 amina amina      4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina    488679 Jan 10 09:32 syslog_systemd.txt
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina      4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Videos
amina@2023-bse-007:~$
```

Step 4: Check if a directory exists

1. Vim append:

   # Directory check

   if [-d "dir1"]; then

     echo "Directory dir1 exists."

   else

     echo "Directory dir1 does not exist. Creating..."

     mkdir -p "dir1"

     echo "Directory dir1 created."

   Fi

```
#!/bin/bash
#Define and show var1
var1="Hello from Lab6"
echo "var1: $var1"
# save ls -l to variable and display
allfiles="$(ls -l)"
echo "allfiles (ls -l ):"
echo "$allfiles"
# directory check
if [-d "dir1"]; then
        echo "directory dir1 exists."
else
        echo "directory dir1 doesnt exist.creating ..."
        mkdir -p "dir1"
        echo "directory dir1 created"
fi
~
~
```

2. Save & quit

3. Run:

   ./setup.sh

```
"setup.sh" 16L, 352B written
amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l ):
total 115676
-rw-rw-r-- 1 amina amina       312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina       253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Desktop
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina         0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root          0 Dec 26 12:44 key
drwxrwxr-x 4 amina amina      4096 Dec 26 03:59 lab4
drwxrwxr-x 2 amina amina      4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina       352 Jan 10 10:08 setup.sh
drwx------ 3 amina amina      4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina    488679 Jan 10 09:32 syslog_systemd.txt
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina      4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina      4096 Dec 26 06:41 Videos
./setup.sh: line 10: [-d: command not found
directory dir1 doesnt exist.creating ...
directory dir1 created
amina@2023-bse-007:~$
```

Step 5: Check if a file exists

1. Vim append:

   # File check

   if [-f "dir1/file2"]; then

     echo "file2 already exists."

else

echo "file2 does not exist. Creating..."

touch "dir1/file2"

chmod a-rwx "dir1/file2"

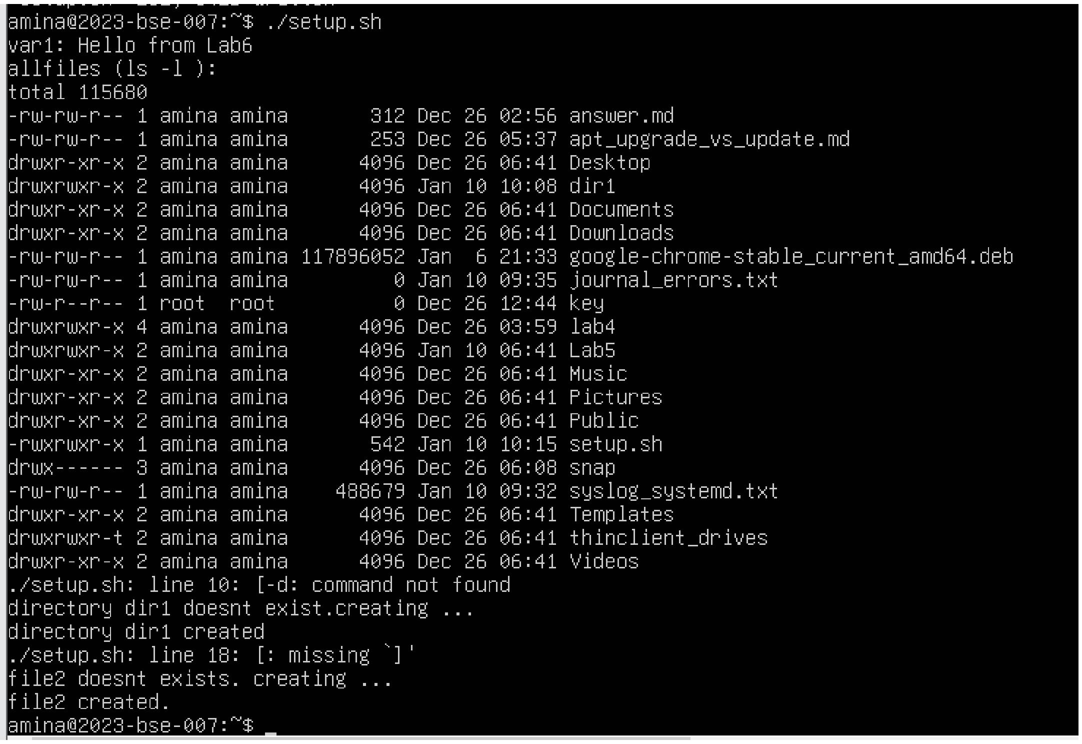
echo "file2 created."

Fi

```
#Define and show var1
var1="Hello from Lab6"
echo "var1: $var1"
# save ls -l to variable and display
allfiles="$(ls -l)"
echo "allfiles (ls -l ):"
echo "$allfiles"
# directory check
if [-d "dir1"]; then
        echo "directory dir1 exists."
else
        echo "directory dir1 doesnt exist.creating ..."
        mkdir -p "dir1"
        echo "directory dir1 created"
fi
# file check
if [ -f "dir1/file2"]; then
        echo "file2 already exits"
else
        echo "file2 doesnt exists. creating ..."
        touch "dir1/file2"
        chmod a-rwx "dir1/file2"
        echo "file2 created."
fi
```

2. Save & quit
3. Run:

./setup.sh



Step 6: Check and grant user permissions

1. Vim append:

    # Permission checks for dir1/file2 (user permissions)

    f="dir1/file2"

    if [! -r "$f"]; then

      echo "Read permission missing; granting to user..."

      chmod u+r "$f"

    fi

    if [! -w "$f"]; then

      echo "Write permission missing; granting to user..."

      chmod u+w "$f"

    fi

    if [! -x "$f"]; then

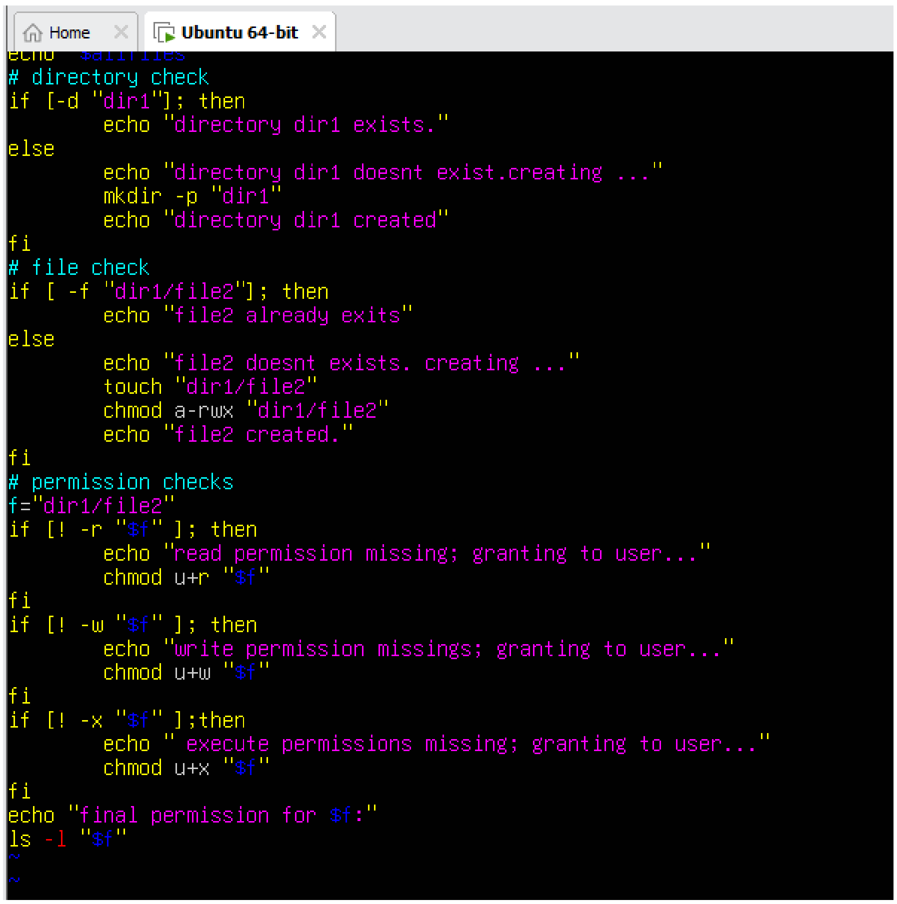
      echo "Execute permission missing; granting to user..."

chmod u+x "$f"

fi

echo "Final permissions for $f:"

ls -l "$f"

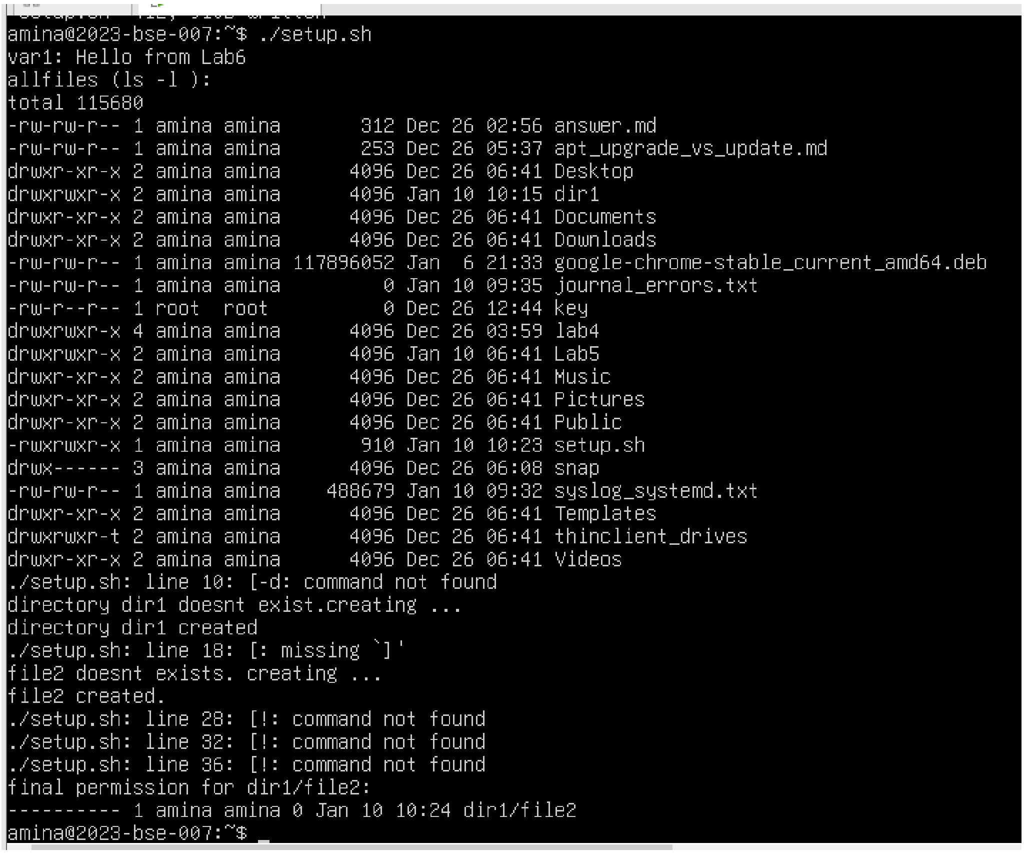



```
# directory check
if [-d "dir1"]; then
        echo "directory dir1 exists."
else
        echo "directory dir1 doesnt exist.creating ..."
        mkdir -p "dir1"
        echo "directory dir1 created"
fi
# file check
if [ -f "dir1/file2"]; then
        echo "file2 already exits"
else
        echo "file2 doesnt exists. creating ..."
        touch "dir1/file2"
        chmod a-rwx "dir1/file2"
        echo "file2 created."
fi
# permission checks
f="dir1/file2"
if [! -r "$f" ]; then
        echo "read permission missing; granting to user..."
        chmod u+r "$f"
fi
if [! -w "$f" ]; then
        echo "write permission missings; granting to user..."
        chmod u+w "$f"
fi
if [! -x "$f" ];then
        echo " execute permissions missing; granting to user..."
        chmod u+x "$f"
fi
echo "final permission for $f:"
ls -l "$f"
~
~
```

2. Save & quit

3. Run:

./setup.sh



Task 11 – Script setup.sh – argument comparisons (eq, ne, gt, lt, ge, le) and string checks

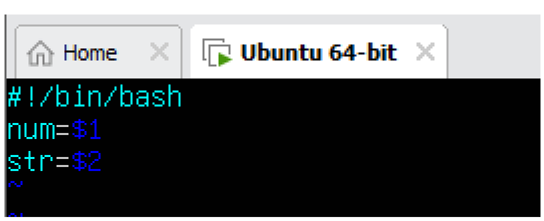
Step 0 – Create script and variables

1. Open vim setup.sh and insert:

#!/bin/bash

num=$1

str=$2



2. Save and quit: :wq

3. Make executable astrnd run example:

chmod +x setup.sh

./setup.sh 10 Student

Step 1 – Numeric equality (-eq)

Append to setup.sh:

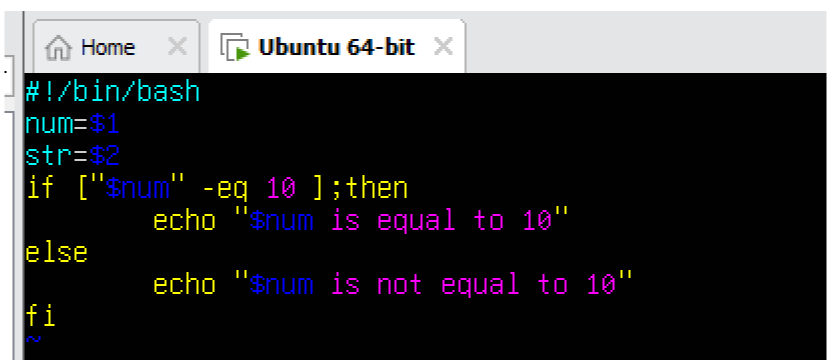
if ["$num" -eq 10]; then

  echo "$num is equal to 10 (-eq)."

else

  echo "$num is NOT equal to 10 (-eq)."

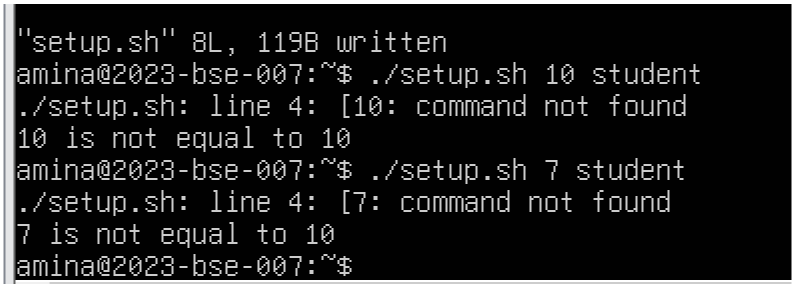
Fi



Run both examples:

./setup.sh 10 Student

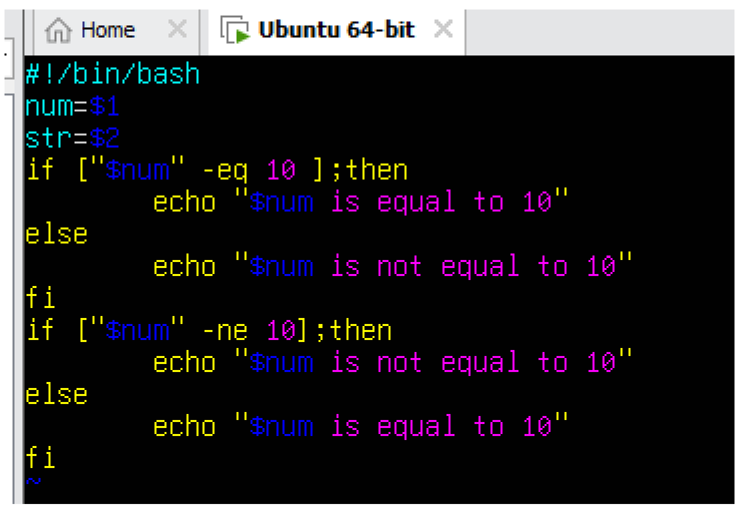
./setup.sh 7 Student



Step 2 – Numeric not equal (-ne)

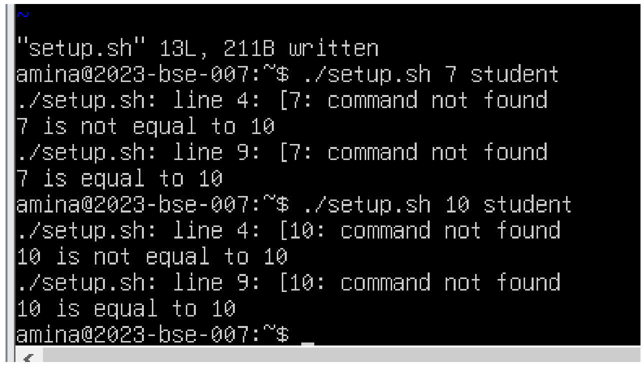
Append:

```
if [ "$num" -ne 10 ]; then
  echo "$num is not equal to 10 (-ne)."
else
  echo "$num is equal to 10 (-ne false)."
fi
```



Run:

```
./setup.sh 7 Student
./setup.sh 10 Student
```



```
~
"setup.sh" 13L, 211B written
amina@2023-bse-007:~$ ./setup.sh 7 student
./setup.sh: line 4: [7: command not found
7 is not equal to 10
./setup.sh: line 9: [7: command not found
7 is equal to 10
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
amina@2023-bse-007:~$ _
```

Step 3 – Greater than (-gt)

Append:

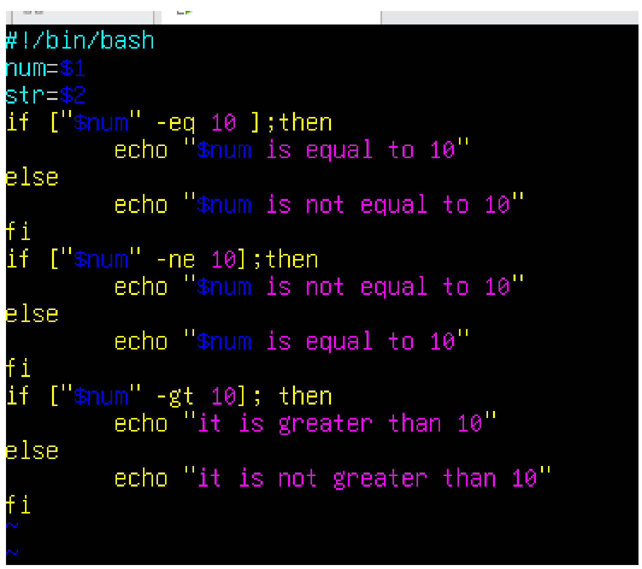
if ["$num" -gt 10]; then

  echo "$num is greater than 10 (-gt)."

else

  echo "$num is NOT greater than 10 (-gt)."

fi

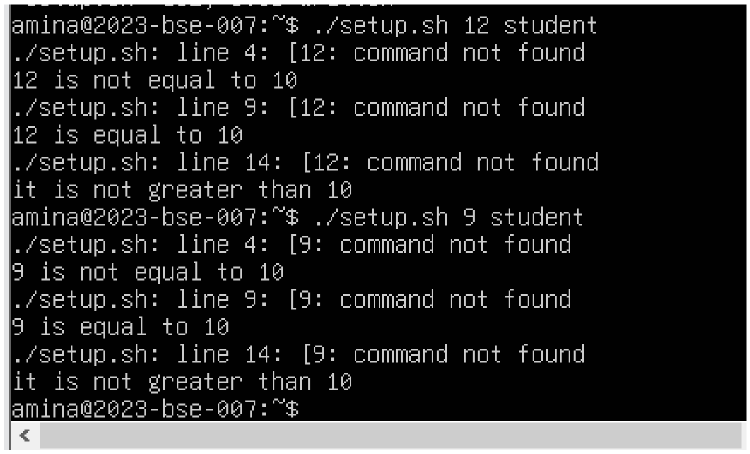


```
#!/bin/bash
num=$1
str=$2
if ["$num" -eq 10 ];then
        echo "$num is equal to 10"
else
        echo "$num is not equal to 10"
fi
if ["$num" -ne 10];then
        echo "$num is not equal to 10"
else
        echo "$num is equal to 10"
fi
if ["$num" -gt 10]; then
        echo "it is greater than 10"
else
        echo "it is not greater than 10"
fi
~
~
```

Run:

./setup.sh 12 Student

./setup.sh 9 Student



```
amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
amina@2023-bse-007:~$ ./setup.sh 9 student
./setup.sh: line 4: [9: command not found
9 is not equal to 10
./setup.sh: line 9: [9: command not found
9 is equal to 10
./setup.sh: line 14: [9: command not found
it is not greater than 10
amina@2023-bse-007:~$
```

Step 4 – Less than (-lt)

Append:

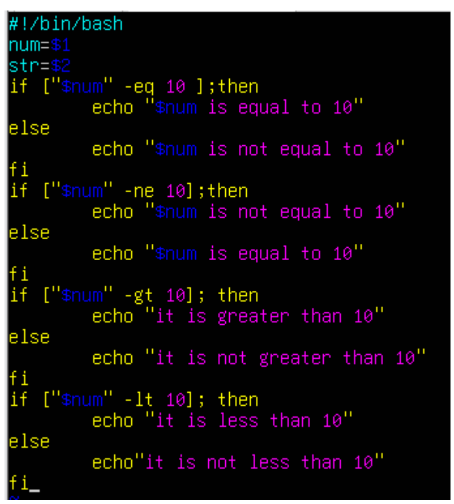
if ["$num" -lt 10]; then

  echo "$num is less than 10 (-lt)."

else

  echo "$num is NOT less than 10 (-lt)."

fi

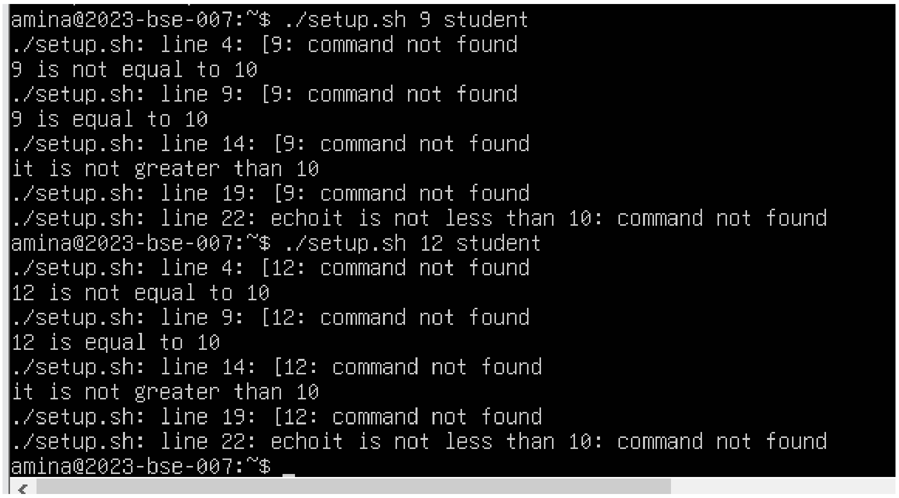


```
#!/bin/bash
num=$1
str=$2
if ["$num" -eq 10 ];then
        echo "$num is equal to 10"
else
        echo "$num is not equal to 10"
fi
if ["$num" -ne 10];then
        echo "$num is not equal to 10"
else
        echo "$num is equal to 10"
fi
if ["$num" -gt 10]; then
        echo "it is greater than 10"
else
        echo "it is not greater than 10"
fi
if ["$num" -lt 10]; then
        echo "it is less than 10"
else
        echo"it is not less than 10"
fi
```

Run:

./setup.sh 9 Student

./setup.sh 12 Student



```
amina@2023-bse-007:~$ ./setup.sh 9 student
./setup.sh: line 4: [9: command not found
9 is not equal to 10
./setup.sh: line 9: [9: command not found
9 is equal to 10
./setup.sh: line 14: [9: command not found
it is not greater than 10
./setup.sh: line 19: [9: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
./setup.sh: line 19: [12: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
amina@2023-bse-007:~$
```

Step 5 – Greater than or equal (-ge)

Append:

```
if [ "$num" -ge 10 ]; then
  echo "$num is greater than or equal to 10 (-ge)."
else
  echo "$num is NOT greater than or equal to 10 (-ge)."
fi
```

```
#!/bin/bash
num=$1
str=$2
if ["$num" -eq 10 ];then
        echo "$num is equal to 10"
else
        echo "$num is not equal to 10"
fi
if ["$num" -ne 10];then
        echo "$num is not equal to 10"
else
        echo "$num is equal to 10"
fi
if ["$num" -gt 10]; then
        echo "it is greater than 10"
else
        echo "it is not greater than 10"
fi
if ["$num" -lt 10]; then
        echo "it is less than 10"
else
        echo"it is not less than 10"
fi
if ["$num" -ge 10]; then
        echo "greater than or equal to 10"
else
        echo "not greater than or equal to 10"
fi
```

Run:

./setup.sh 10 Student

./setup.sh 8 Student

```
"setup.sh" 28L, 507B written
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [10: command not found
not greater than or equal to 10
amina@2023-bse-007:~$ ./setup.sh 8 student
./setup.sh: line 4: [8: command not found
8 is not equal to 10
./setup.sh: line 9: [8: command not found
8 is equal to 10
./setup.sh: line 14: [8: command not found
it is not greater than 10
./setup.sh: line 19: [8: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [8: command not found
not greater than or equal to 10
amina@2023-bse-007:~$
```

Step 6 – Less than or equal (-le)

Append:

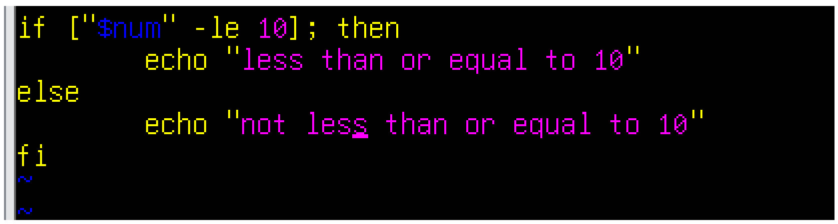
if ["$num" -le 10]; then

  echo "$num is less than or equal to 10 (-le)."

else

  echo "$num is NOT less than or equal to 10 (-le)."

fi

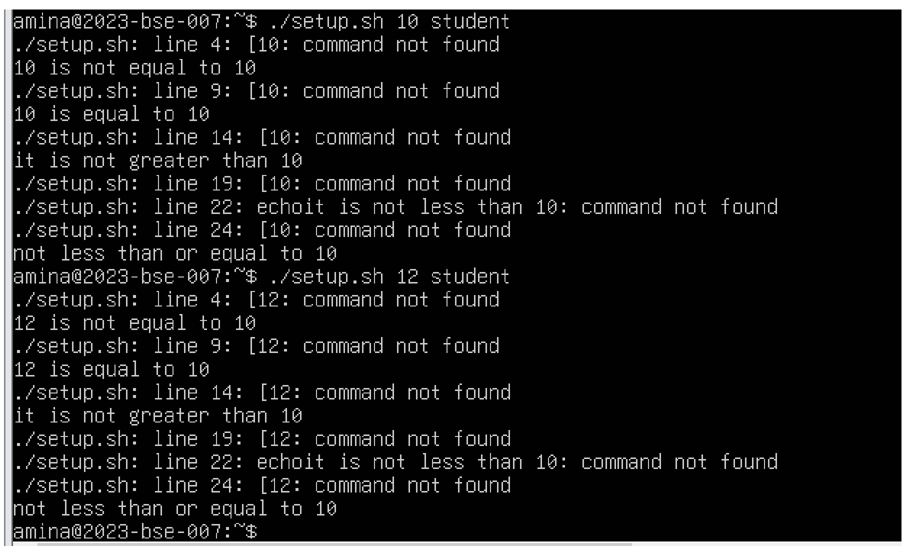


```
if ["$num" -le 10]; then
        echo "less than or equal to 10"
else
        echo "not less than or equal to 10"
fi
~
~
```

Run:

./setup.sh 10 Student

./setup.sh 12 Student



```
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [10: command not found
not less than or equal to 10
amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
./setup.sh: line 19: [12: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [12: command not found
not less than or equal to 10
amina@2023-bse-007:~$
```

Step 7 – String equality (=)

Append:

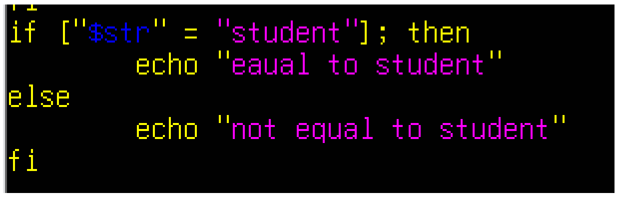
if ["$str" = "Student"]; then

  echo "Second argument equals 'Student' (=)."

else

  echo "Second argument does NOT equal 'Student' (=)."

fi

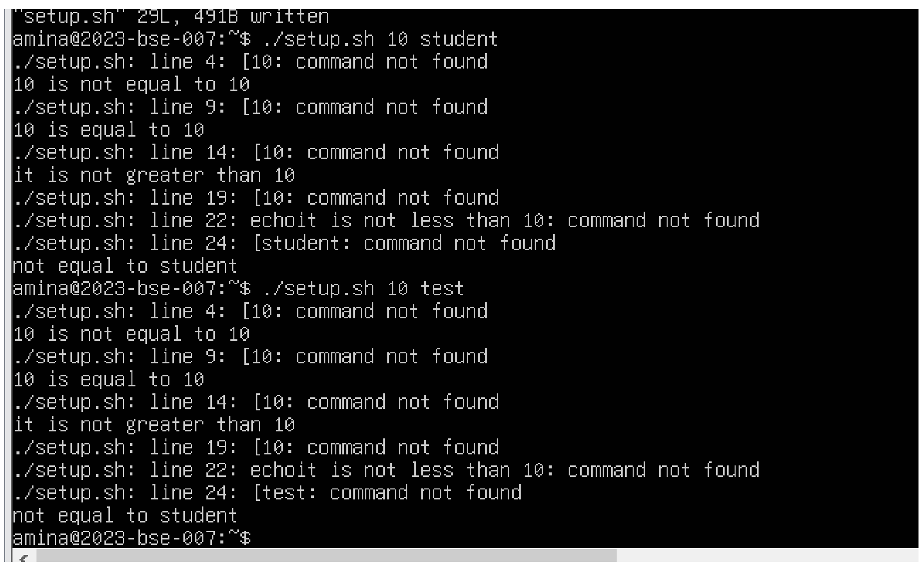


```
if ["$str" = "student"]; then
        echo "eaual to student"
else
        echo "not equal to student"
fi
```

Run:

./setup.sh 10 Student

./setup.sh 10 Test



```
"setup.sh" 29L, 491B written
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [student: command not found
not equal to student
amina@2023-bse-007:~$ ./setup.sh 10 test
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [test: command not found
not equal to student
amina@2023-bse-007:~$
```

Step 8 – String inequality (!=)

Append:

if ["$str" != "Student"]; then

  echo "Second argument is not equal to 'Student' (!=)."

else

  echo "Second argument equals 'Student' (!= false)."

fi

```
if ["$str" != "student"]; then
        echo "not equal to student"
else
        echo "equal to student"
fi
```

Run:

./setup.sh 10 Test

./setup.sh 10 Student

```
"setup.sh" 29L, 492B written
amina@2023-bse-007:~$ ./setup.sh 10 test
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [test: command not found
equal to student
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [student: command not found
equal to student
amina@2023-bse-007:~$ _
```

Step 9 – Check if second argument is empty (-z)

Append:

if [-z "$str"]; then

  echo "Second argument is empty (zero-length)."

else

  echo "Second argument is not empty."

fi

```
if [-z "$str" ]; then
        echo "empty"
else
        echo "not empty"
fi
```

Run:

./setup.sh 10

./setup.sh 10 Student

```
"setup.sh" 29L, 461B written
amina@2023-bse-007:~$ ./setup.sh 10
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [-z: command not found
not empty
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [-z: command not found
not empty
amina@2023-bse-007:~$ _
```

Task 12 – Script setup.sh – print all arguments with a for loop

Step 1 – Create script with shebang and comment

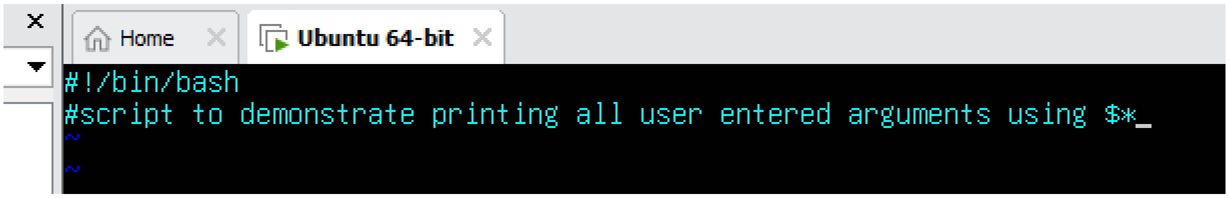
1. Open vim and overwrite setup.sh:

vim setup.sh

2. Insert the following lines:

#!/bin/bash

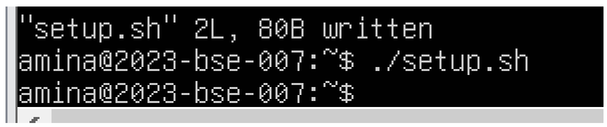
Script to demonstrate printing all user-entered arguments using $*



3. Save and quit: :wq

4. Run the script (no arguments expected yet):

./setup.sh



Step 2 – Append for loop to print all arguments

1. Re-open setup.sh in vim:

vim setup.sh

2. Append the following lines **below the comment**:

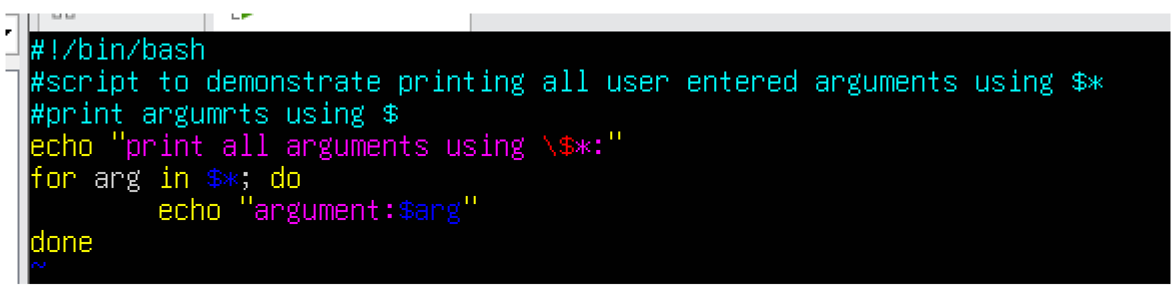
Print all arguments using $*

echo "Printing all arguments using \$*:"

for arg in $*; do

  echo "Argument: $arg"

done



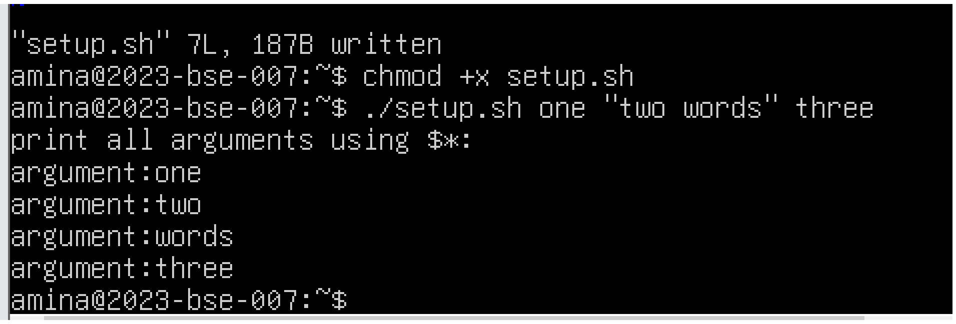
3. Save and quit: :wq

4. Make the script executable (if not already):

chmod +x setup.sh

5. Run the script with example arguments:

./setup.sh one "two words" three



Task 13 – Script setup.sh – while loop summation and functions

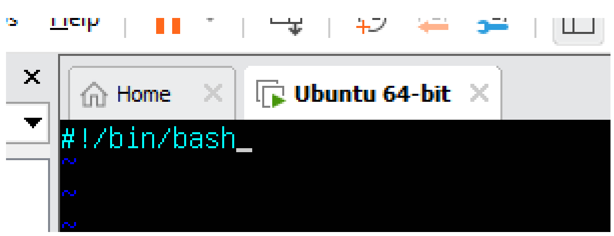
Step 1 – Create script with shebang

1. Open vim and overwrite setup.sh:

vim setup.sh

2. Insert:

#!/bin/bash

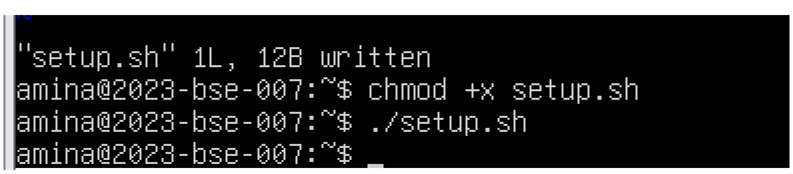


3. Save and quit: :wq

4. Make executable and run (no output expected):

chmod +x setup.sh

./setup.sh



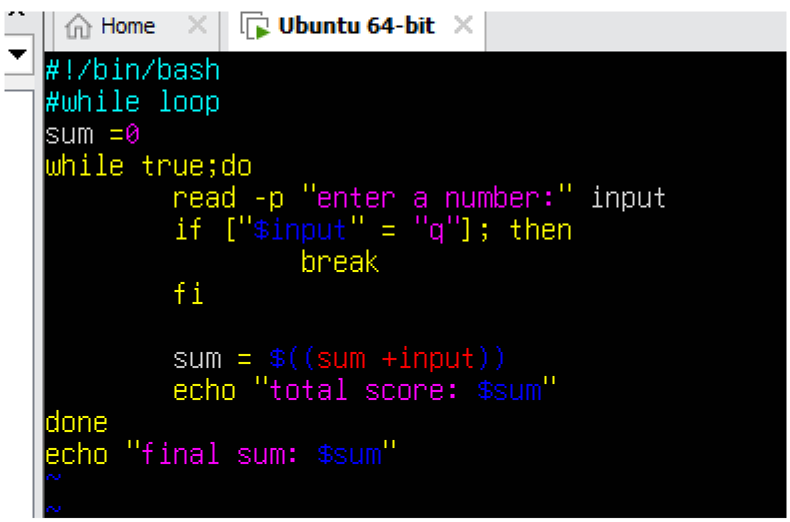
Step 2 – Add interactive while-loop summation

1. Re-open setup.sh in vim:

vim setup.sh

2. Append the following:

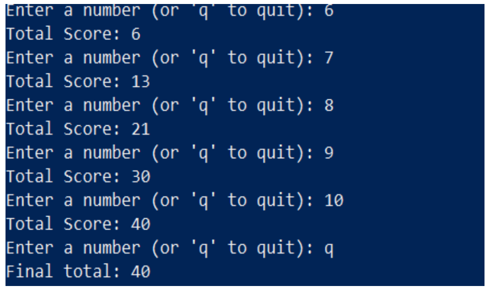
```
# While-loop summation (interactive)
sum=0
while true; do
  read -p "Enter a number (or 'q' to quit): " input
  if [ "$input" = "q" ]; then
    break
  fi
  sum=$((sum + input))
  echo "Total Score: $sum"
done
echo "Final total: $sum"
```



3. Save and quit: :wq

4. Run the script and demonstrate a short session:

./setup.sh

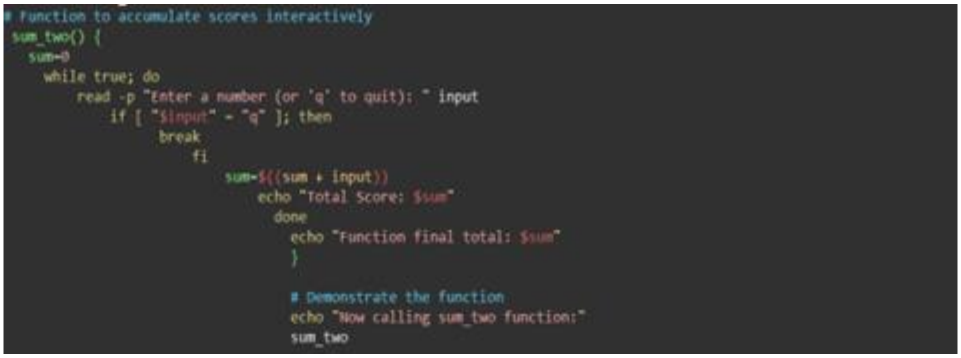


```
Enter a number (or 'q' to quit): 6
Total Score: 6
Enter a number (or 'q' to quit): 7
Total Score: 13
Enter a number (or 'q' to quit): 8
Total Score: 21
Enter a number (or 'q' to quit): 9
Total Score: 30
Enter a number (or 'q' to quit): 10
Total Score: 40
Enter a number (or 'q' to quit): q
Final total: 40
```

Step 3 – Move interactive while-loop into a function sum_two()

1. Re-open setup.sh in vim.
2. Replace/delete the standalone while-loop **if you want the function to handle input only**, then append:

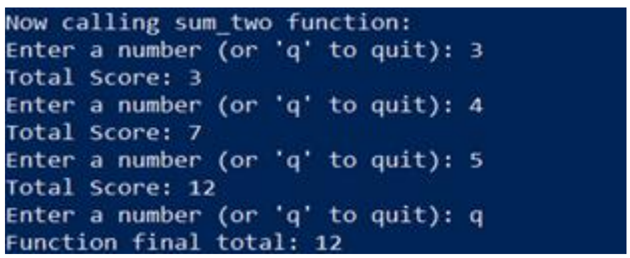
```
# Function to accumulate scores interactively

sum_two() {

  sum=0

  while true; do

    read -p "Enter a number (or 'q' to quit): " input

    if [ "$input" = "q" ]; then

      break

    fi

    sum=$((sum + input))

    echo "Total Score: $sum"

  done

  echo "Function final total: $sum"

}

# Demonstrate the function

echo "Now calling sum_two function:"

sum_two
```



```
# Function to accumulate scores interactively
sum_two() {
  sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
            if [ "$input" = "q" ]; then
                break
                    fi
                        sum=$((sum + input))
                            echo "Total Score: $sum"
                              done
                                echo "Function final total: $sum"
                                }

                                # Demonstrate the function
                                echo "Now calling sum_two function:"
                                sum_two
```

3. Save and quit: :wq
4. Run the script and demonstrate short session for function:

./setup.sh



```
Now calling sum_two function:
Enter a number (or 'q' to quit): 3
Total Score: 3
Enter a number (or 'q' to quit): 4
Total Score: 7
Enter a number (or 'q' to quit): 5
Total Score: 12
Enter a number (or 'q' to quit): q
Function final total: 12
```

Step 4 – Add function that sums two numeric arguments and returns result

1. Re-open setup.sh in vim.
2. Append the following:

Function that sums two arguments and returns the result

sum_args() {

 a=$1

 b=$2

 return $((a + b))

}

Demonstrate sum_args function

echo "Now demonstrating sum_args function:"

sum_args 3 4

result=$?

echo "sum_args(3,4) returned: $result"

```
# Function that sums two arguments and returns the result
sum_args() {
  a=$1
    b=$2
      return $((a + b))
      }

      # Demonstrate sum_args function
      echo "Now demonstrating sum_args function:"
      sum_args 3 4
      result=$?
      echo "sum_args(3,4) returned: $result"
```

3. Save and quit: :wq
4. Run the script and capture the demonstration output:

chmod +x setup.sh

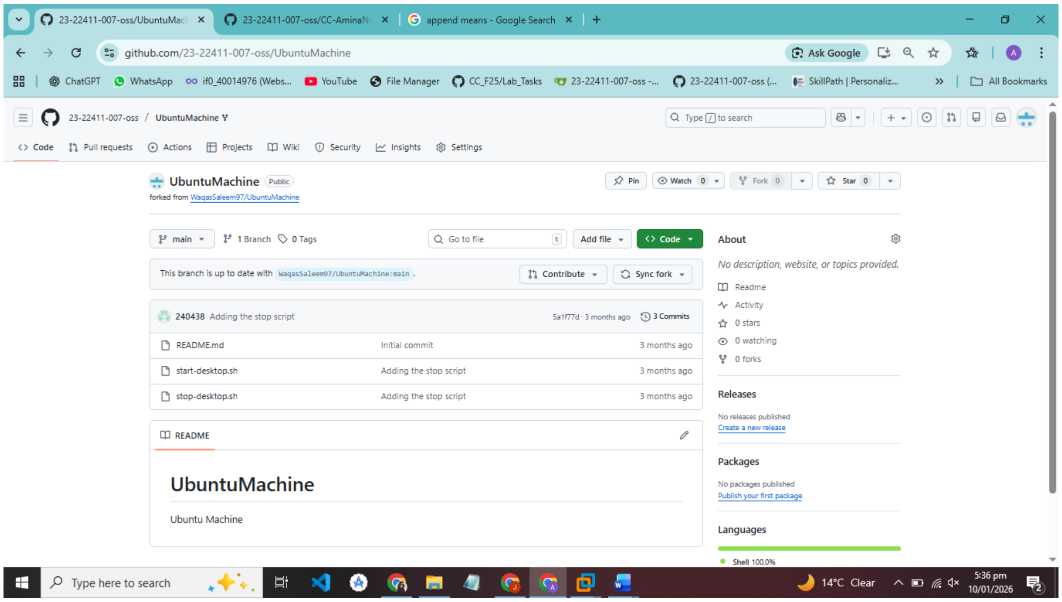
./setup.sh

```
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
```

Task 14 – Codespaces GUI — fork repo, run start-desktop.sh, open VNC, stop GUI

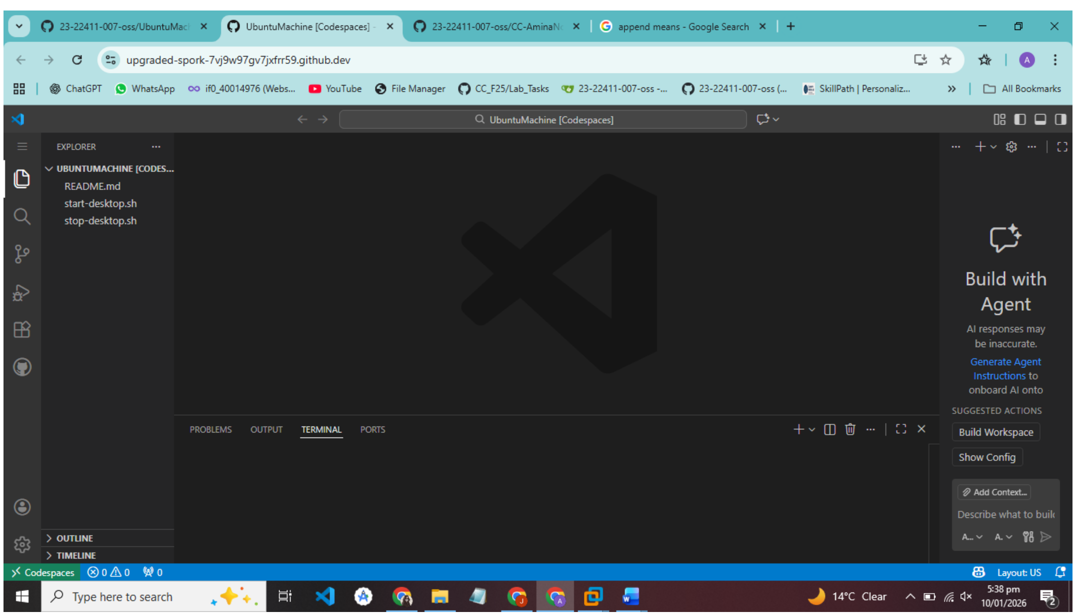
Step 1: Fork the repository to your GitHub account

1. Open the GitHub repo URL in your browser (the repo your instructor gave you).
2. Click the “Fork” button on the top-right corner.
3. Choose your GitHub account as the destination.
4. Wait for the fork to complete.



Step 2: Open a Codespace on your fork

1. Go to your forked repo on GitHub.
2. Click the green “Code” button → select Open with Codespaces → Create codespace on main (or whatever branch).
3. Wait for Codespaces to fully load.



Step 3: Verify the start/stop scripts exist and are executable

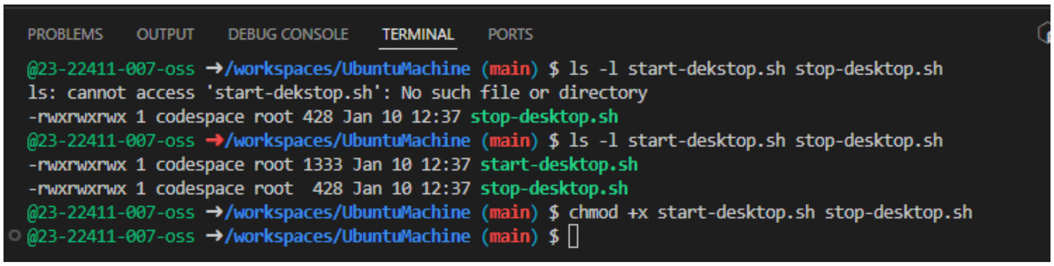
1. Open the Codespace terminal.

2. Check for the scripts:

   ls -l start-desktop.sh stop-desktop.sh

3. If they are not executable, make them executable:

   chmod +x start-desktop.sh stop-desktop.sh

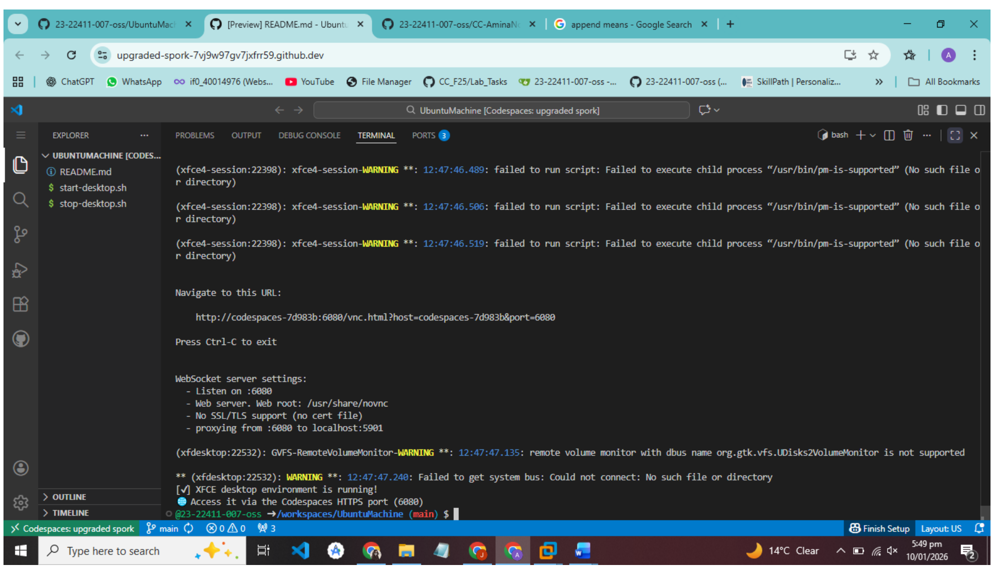


Step 4: Start the desktop GUI

1. In the Codespace terminal, ensure the script is executable and run it:

   ./start-desktop.sh

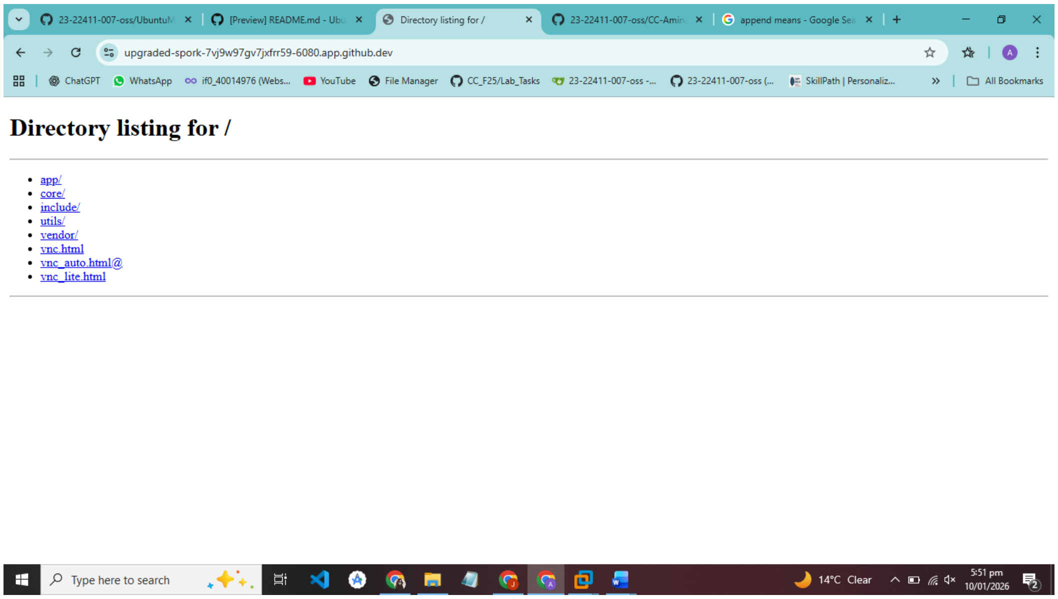
2. Wait for the script to finish; it should display messages like “VNC server started on port 6080”.



Step 5: Verify forwarded ports

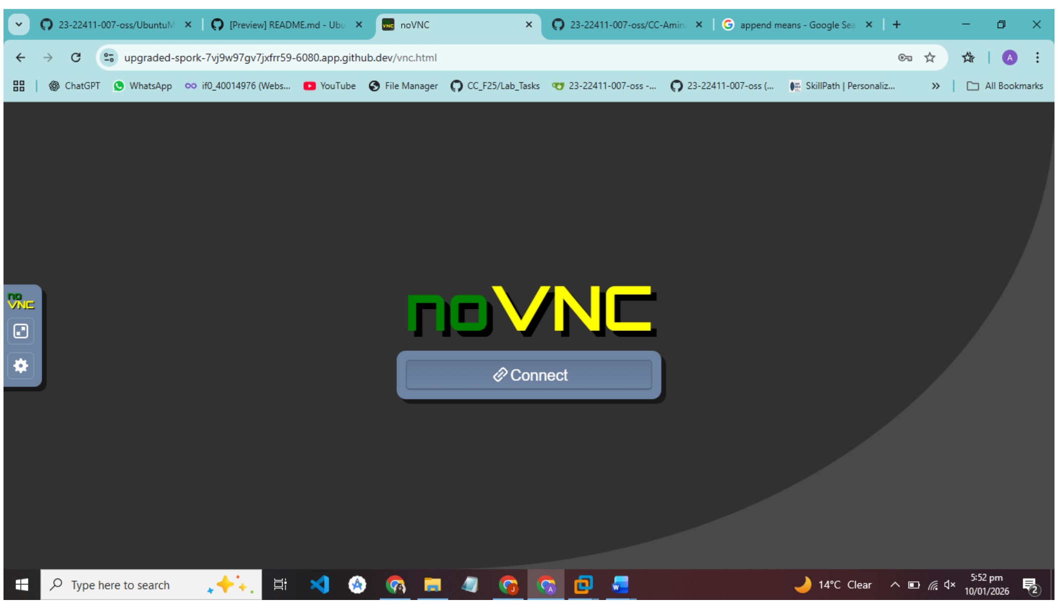
1. In Codespaces, open the Ports view (usually bottom panel or via the sidebar).

2. Confirm port 6080 is listed and forwarded.

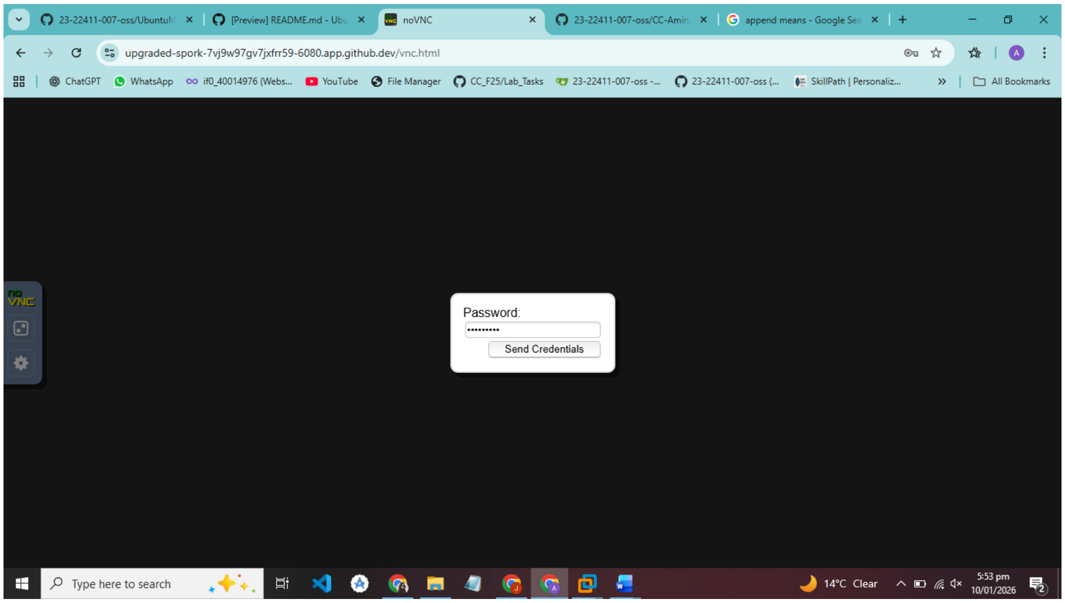


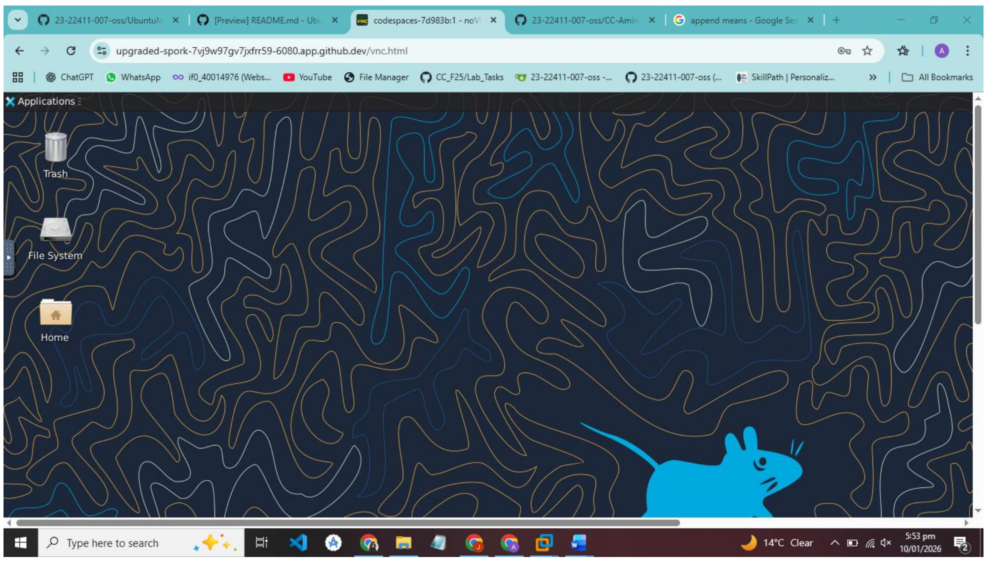
Step 6: Open the VNC HTML page

1. Click the preview URL for port 6080 or copy it into your browser.
2. Open vnc.html at that URL.



3. When prompted for a password, enter: codespace





Step 7: Stop the GUI

1. Go back to Codespace terminal.
2. Run:

3. ./stop-desktop.sh
4. Wait for the script to finish; it should show messages about stopping the VNC/GUI.